Discrete Mathematics and Its Applications

Eighth Edition

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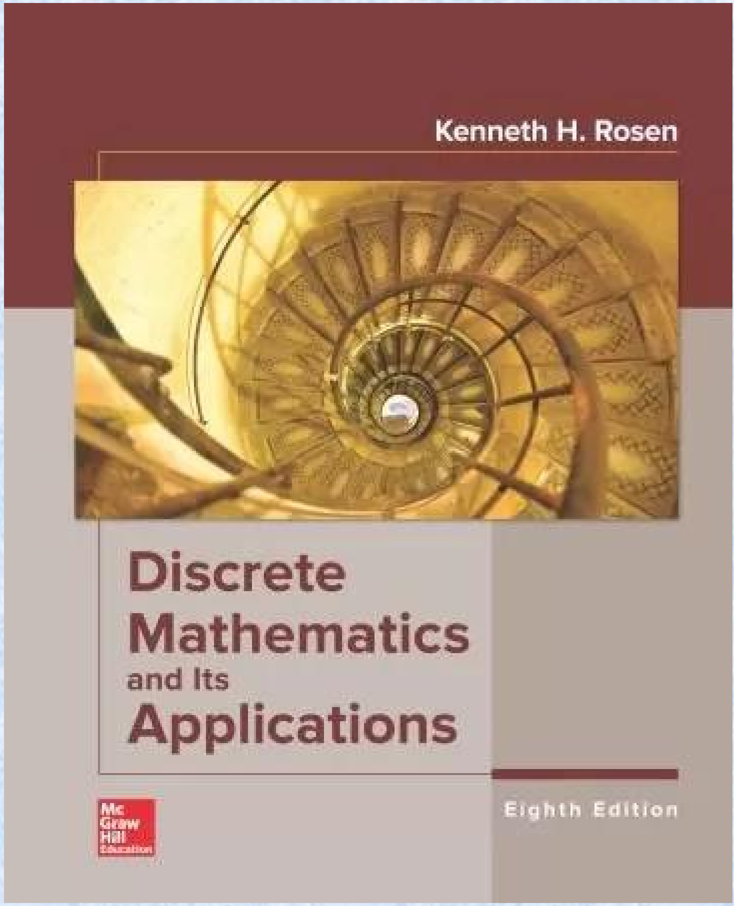


本学期教学安排

- 9 Relations
- 9 Semigroups and Groups P319-356
- 11 Groups and Coding P401-423
- 8 Advanced Counting Techniques
- 10 Graphs
- 11 Trees

Discrete Mathematics

- Logic
  - Propositional Logic
    - Proposition and Connectives
    - Propositional Equivalences
    - Translating
    - disjunctive and conjunctive
      - Normal Form
  - Predicate Logic
    - Predicates and Quantifiers
    - Logical Equivalences Involving Quantifiers
    - Translating
    - Prenex Normal Form
  - Reasoning
    - Rules of Inference for Propositional Logic
    - Rules of Inference for Predicate Logic
    - Proofs
      - Direct Proofs Proof by Contraposition
      - Mathematical Induction &Recursive Definitions
      - Strong Induction Well-Ordering Property
- Set Theory
  - Sets and Operations
  - Cardinality of Sets Countable and Uncountable
  - Functions onto function one-to-one function Growth of Functions Big-O Big-Omega Big-Theta
  - *Relations*
    - Relations and Their Properties reflexive symmetric antisymmetric transitive
    - Closures of Relations Warshall's Algorithm
    - Equivalence Relations Equivalence Classes and Partitions of a Set
    - Partial Orderings Poset Hasse diagram Maximal and Minimal Elements LUB GLB Lattices Topological Sort
- Counting
  - Number Theory Modular Arithmetic Modular Exponentiation Primes GCD LCM Solving Congruences Cryptography
  - Basics of Counting Pigeonhole Principle Permutation and Combinations P and C with Repetition
  - *Discrete Probability*
    - Conditional Probability Independence
    - Sample space Probability of events Probability Theory
    - Random Variables Expected Value and Variance
    - Bayes' Theorem
  - *Recurrence Relations*
    - Applications of Recurrence Relation Modeling with RR
    - Solving Linear Recurrence Relations characteristic equation
    - Divide-and-Conquer Algorithms and RR
- Graph Theory
  - Graph Terminology Kn Wn Cn Qn Kn,m
  - Isomorphic graph invariant Path and connected components(strongly weakly)
  - Determine Euler and Hamilton path and circuit
  - Dijkstra's Algorithm
  - Planar Graphs Euler's Formula Determine Planar Graphs Kuratowski's Theorem
  - Proof Five Color Theorem Chromatic number of graphs Chromatic Polynomial
  - Labeling algorithm Max Flow Min Cut Theorem
  - Tree
    - Properties of trees Rooted tree
    - Full m-ary tree Balanced tree Binary search tree
    - Prefix Codes Huffman Coding
    - Traversal algorithm Preorder Inorder Postorder Infix Prefix Postfix Notation
    - Depth-First Search Breadth-First Search Backtracking
    - Prim's algorithm Kruskal's algorithm
- Group Theory
  - Mathematical structure Algebraic system Binary operation Closure Associative Distributive De Morgan's laws Identity Inverse
  - Semigroup Monoid Group Abelian group finite group Order of group S3 Zn
  - Isomorphism and Homomorphism Congruence relation Fundamental Homomorphism Theorem
  - Product algebra Quotient algebra Sub algebra Normal subgroup Kernel of f
  - Coding
    - Encoding function Detect errors Group Codes mod-2 Boolean product Parity check matrix
    - Decoding table Correct errors Maximum likelihood decoding function Syndrome

Relations
关系

Rosen 8th ed., ch. 9

9 Relations

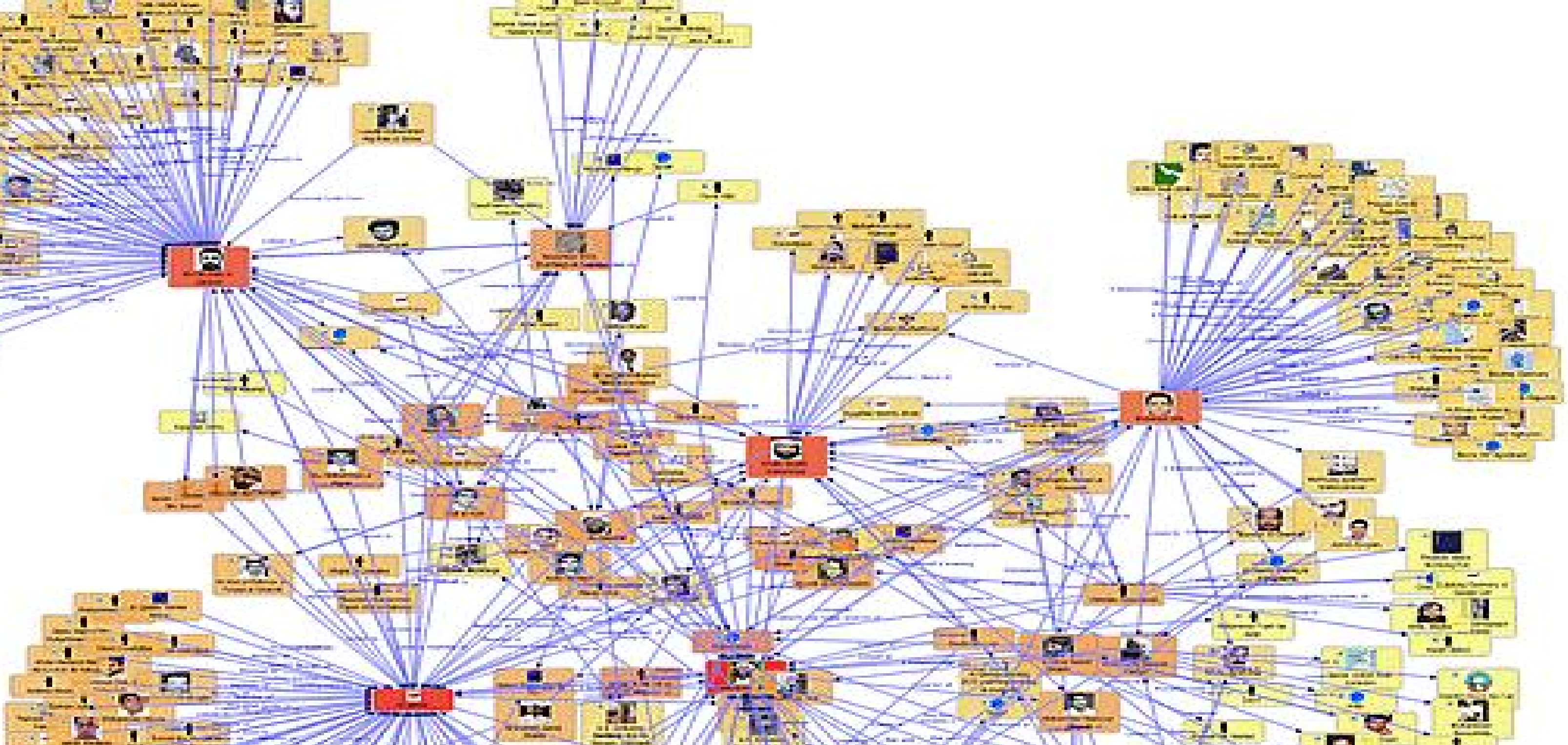
- 9.1 Relations and Their Properties
- 关系及关系性质
- 9.2 *n*-ary Relations and Their Applications
- n元关系及应用
- 9.3 Representing Relations 关系的表示
- 9.4 Closures of Relations 关系闭包
- 9.5 Equivalence Relations 等价关系
- 9.6 Partial Orderings 偏序关系

9.1 Relations and Their Properties

- Introduction
- Functions as relations
- Relations on Set
- Properties of Relations
- Combining Relations

What are relations?

- Relations are a formal means to specify which elements from two or more sets are related to each other
- Examples
  - {students} who take {courses}
  - {businesses} and their {telephone numbers}
  - {integers} and their {divisors}
  - {program variables} and the {subroutines} they are used in



美国击毙"基地组织"头目细节曝光：现身阳台，被地狱火导弹绞碎

2022-08-03 18:03

据外媒报道，当地时间8月1日，美国总统拜登宣布，美国在阿富汗首都喀布尔发动无人机空袭，击毙了"基地"组织头目艾曼·扎瓦希里。扎瓦希里是美国9·11恐怖袭击的主要策划者。

Network Metrics

Calculate | CardView | TableView | Group area | Expand groups | Collapse groups

Name	Type	Degree	Betweenness	Closeness	Eigenvalue	Hub	Authority
Osama bin Laden	Person	44	0.920492092358...	1	0.0271	0	0.011
Abdallah Al-Halabi	Person	2	0	0.654867256637...	0.0001	0	0
Abu Mussab al-Zarqawi	Person	84	0.934887847326...	0.869451697127...	0.7028	0.6572	0.1076
Al Qaeda	Terrorist Organiz...	85	1	0.962427745664...	0.0416	0.3941	0.0156
Ayman Al-Zawahiri	Person	14	0.045794908783...	0.716129032258...	0	0	0.0173
Enaam Arnaout	Person	4	0.031189325814...	0.656804733727...	0.0001	0	0
Imad Eddin Barakat Yarbas	Person	11	0.065049589038...	0.704016913319...	0.0015	0	0.0025
Khalid Shaikh Mohammed	Person	32	0.339916464724...	0.866059817945...	0.002	0	0.1528
Mohamed Atta	Person	61	0.666268740074...	0.820197044334...	0.0015	0	0.6816

Definition of relations

- Let A and B be two sets. A *binary relation* R *from* A *to* B is a subset of $A \times B$.
  - Note that the order of the two sets matters.
- More generally, let A_1, A_2, ..., A_n be n sets. An *n-ary relation* R on these sets is a subset of $A_1 \times A_2 \times \ldots \times A_n$.
  - The sets A_i are known as the *domains* of the relation, and n as its *degree*
  - Again, the order of the domains matters.

Binary Relations

- Let A, B be any sets. A *binary relation* R from A to B, written (with signature) $R{:}A\times B$, or $R{:}A,B$, is (can be identified with) a subset of the set $A\times B$.
  - *E.g.*, let $< \,:\, \mathbf{N}\leftrightarrow\mathbf{N} :\equiv \{(n,m) \mid n < m\}$
- The notation $a\ R\ b$ or aRb means that $(a,b)\in R$.
  - *E.g.*, $a < b$ means $(a,b)\in <$
- If aRb we may say "*a* is related to *b* (by relation *R*)",
  - or just "*a* relates to *b* (under relation *R*)".
- A binary relation R corresponds to a predicate function $P_R{:}A\times B\rightarrow\{\mathbf{T},\mathbf{F}\}$ defined over the 2 sets A,B;
  - *e.g.*, predicate "eats" $:\equiv \{(a,b) \mid \text{organism } a \text{ eats food } b\}$

定义： 二元关系/binary relation

-----按照某种规定，定义了一个有序对（a， b）的集合R，其中 $a\in A$， $b\in B$，称R为A到B的一个二元关系。

Let A and B be sets. A binary relation from A to B is a subset of A×B.

$$R=\{(a, b) \mid a\in A,\ b\in B,\ a\ r\ b\}$$

其中 a r b 读作 a 相关于 b 。

注意： 二元关系是指满足某规律的有序对的全体。

Complementary Relations补关系

- Let R:A,B be any binary relation.
- Then, $\cancel{R}$:$A \times B$, the *complement* of R, is the binary relation defined by
  $$\cancel{R} :\equiv \{(a,b) \mid (a,b) \notin R\} = (A \times B) - R$$
- Note this is just $\overline{R}$ if the universe of discourse is $U = A \times B$; thus the name *complement.*
- Note the complement of $\cancel{R}$ is R.

Example: $\not< \; = \{(a,b) \mid (a,b) \notin \; <\} = \{(a,b) \mid \neg a < b\} = \; \geq$

Inverse Relations 逆关系

- Any binary relation $R{:}A\times B$ has an *inverse* relation $R^{-1}{:}B\times A$, defined by

  $R^{-1} :\equiv \{(b,a) \mid (a,b)\in R\}$.

  E.g., $<^{-1} = \{(b,a) \mid a<b\} = \{(b,a) \mid b>a\} = >$.
- *E.g.*, if R:People→Foods is defined by

  $a\ R\ b \Leftrightarrow a \text{ eats } b$, then:

  $b\ R^{-1}\ a \Leftrightarrow b \text{ is eaten by } a$. (Passive voice.)

Functions

- A function $f : A \rightarrow B$ is a special case of a relation from A to B
  - It satisfies the property: for each $x \in A$, there is exactly one pair $(x, f(x))$ in the relation
- Relations are a generalization of functions
  - They allow unmapped elements from A
  - They allow one-to-many mappings where an element from A maps to multiple elements from B
  - They also allow many-to-one and many-to-many mappings

Relations on a set

- A *relation R on a set A* is a relation from A to A, that is, a subset of $A^2 = A \times A$.
- Examples (on the set of integers):
  - x is equal to y
  - x is greater than y
  - x divides y
- Note that in these cases both A and the relation R are infinite sets

Relations on a Set

- A (binary) relation from a set *A* to itself is called a relation *on* the set *A*.
- *E.g.*, the “<” relation from earlier was defined as a relation *on* the set **N** of natural numbers.
- The (binary) *identity relation* $\mathbf{I}_A$ on a set *A* is the set $\{(a,a)|a\in A\}$.

Example 4

- Let A be the set {1,2,3,4}, Which ordered pairs are in the relation R={(a,b) | a divides b}?

- {(1,1),(2,2),(3,3),(4,4),(1,2),(1,3),(1,4),(2,4)}

Example 5

- Consider these relations on the set of integers:
- $R_1=\{(a,b) \mid a \leq b\}$,
- $R_2=\{(a,b) \mid a>b\}$,
- $R_3=\{(a,b) \mid a=b \text{ or } a=-b\}$,
- $R_4=\{(a,b) \mid a=b\}$,
- $R_5=\{(a,b) \mid a=b+1\}$,
- $R_6=\{(a,b) \mid a+ b \leq 3\}$,
- Which of these relations contain each of the pairs (1,1),(1,2),(2,1),(1,-1) and (2,2)?

How many relations are there?

- Relations on a set A with n elements
- A^2 has n^2 elements. We know that the number of subsets of a set with m elements is 2^m, so the desired number is 2^{n^2}.
- The number of functions from A to A is n^n
- Is this asymptotically the same?
- $n^n = (2^{\log n})^n = 2^{n \log n}$ which grows slower than 2^{n^2}.

Properties of Relations

- Reflexive 自反
- Symmetric 对称
- Transitive 传递

Reflexivity

- A relation R on A is *reflexive* if $\forall a \in A,\ aRa$.
  - *E.g.*, the relation $\geq \;:\equiv \{(a,b) \mid a \geq b\}$ is reflexive.
- A relation R is *irreflexive* iff its *complementary* relation $\not R$ is reflexive.
  - Example: **<** is irreflexive, because ≥ is reflexive.
  - Note "*irreflexive*" does **NOT** mean "*not reflexive*"!
    - For example: the relation "likes" between people is not reflexive, but it is not irreflexive either.
      - Since not everyone likes themselves, but not everyone *dislikes* themselves either!

Example 7

- Consider these relations on {1,2,3,4}:
- R_1={(1,1),(1,2),(2,1),(2,2),(3,4),(4,1),(4,4)},
- R_2={(1,1),(1,2),(2,1)},
- R_3={(1,1),(1,2),(1,4),(2,1),(2,2),(3,3),(4,1),(4,4)},
- R_4={(2,1),(3,1),(3,2),(4,1),(4,2),(4,3)},
- R_5={(1,1),(1,2),(1,3),(1,4),(2,2),(2,3),(2,4),(3,3),(3,4),(4,4)},
- R_6={(3,4)},
- Which of these relations are reflexive?

Example 8,9

- Which of the relations from Example 5 are reflexive?

- Is the “divides” relation on the set of positive integers reflexive?

Symmetry & Antisymmetry

- A binary relation R on A is *symmetric* iff $R = R^{-1}$, that is, if $(a,b)\in R \leftrightarrow (b,a)\in R$.
  - *E.g.*, **=** (equality) is symmetric. **<** is not.
  - "is married to" is symmetric, "likes" is not.
- A binary relation R is *antisymmetric* if $\forall a\neq b, (a,b)\in R \rightarrow (b,a)\notin R$.
  - **Examples: <** is antisymmetric, "likes" is not.
  - **Exercise:** prove this definition of antisymmetric is equivalent to the statement that $R\cap R^{-1} \subseteq \mathbf{I}_A$.

Example 10 11 12

- Relations in Example 7
- Relations in Example 5
- “divides” relation

- Symmetry & Antisymmetry?

- *Asymmetric* 非对称 $(a,b) \in R \rightarrow (b,a) \notin R$

Transitivity

- A relation R is *transitive* iff (for all a,b,c)
  $(a,b)\in R \land (b,c)\in R \rightarrow (a,c)\in R$.
- A relation is *intransitive* iff it is not transitive.
- Some examples:
  - "is an ancestor of" is transitive.
  - "likes" between people is intransitive.
  - "is located within 1 mile of" is… ?

Example 14 15 16

- Relations in Example 7
- Relations in Example 5
- “divides” relation

- Transitivity?

Examples in Mathematics

- $=$ (equality)
- $\neq$ (inequality)
- $\emptyset$ (empty relation)

Properties of relations

- A relation R on a set A is
- *reflexive*, if
  - $\forall x \in A, (x, x) \in R$
- *symmetric*, if
  - $\forall x \forall y, (x, y) \in R \Rightarrow (y, x) \in R$
- *antisymmetric*, if
  - $\forall x \forall y, (x, y) \in R$ and $(y, x) \in R \Rightarrow x = y$
- *transitive*, if
  - $\forall x \forall y \forall z, (x, y) \in R$ and $(y, z) \in R \Rightarrow (x, z) \in R$

Example

- Consider the "divides" relation on the set of nonnegative integers: $x|y$ iff x divides y.
- Reflexive?
  - NO. 0 does not divide 0. (Would be reflexive on the set of positive integers)
- Symmetric?
  - NO. 2 divides 4, but not the other way around.
- Antisymmetric? YES
- Transitive? YES

Counting the reflexive relations

- How many reflexive relations are there on a set *A* with *n* elements?

Counting the reflexive relations

- A relation R on a set A is a subset of $A \times A$. Consequently, a relation is determined by specifying whether each of the n^2 ordered pairs in $A \times A$ is in R. However, if R is reflexive, each of the n ordered pairs (a, a) for $a \in A$ must be in R. Each of the other $n(n - 1)$ ordered pairs of the form (a, b), where $a \neq b$, may or may not be in R. Hence, by the product rule for counting,there are $2^{n(n-1)}$ reflexive relations [this is the number of ways to choose whether each element (a, b), with $a \neq b$, belongs to R].

Combining relations

- Since relations are by definition sets, they can be combined in any way that sets can be combined, including union, intersection, and set difference.

Example 17

- Let A={1,2,3} and B={1,2,3,4}. The relations R1={(1,1),(2,2),(3,3)} and R2={(1,1),(1,2),(1,3),(1,4)} can be combined to obtain
- R1∩R2={(1,1)}
- R1UR2={(1,1),(1,2),(1,3),(1,4),(2,2),(3,3)}
- R1-R2 ={(2,2),(3,3)}
- R2-R1 ={(1,2),(1,3),(1,4)}

Example 18 19

- R1 consists of all ordered pair (a,b), where a is a student who has taken course b,
- R2 consists of all ordered pair (a,b),where a is a student who requires course b to graduate.
- What are the relations R1UR2 ,R1∩R2 R1-R2 R2-R1

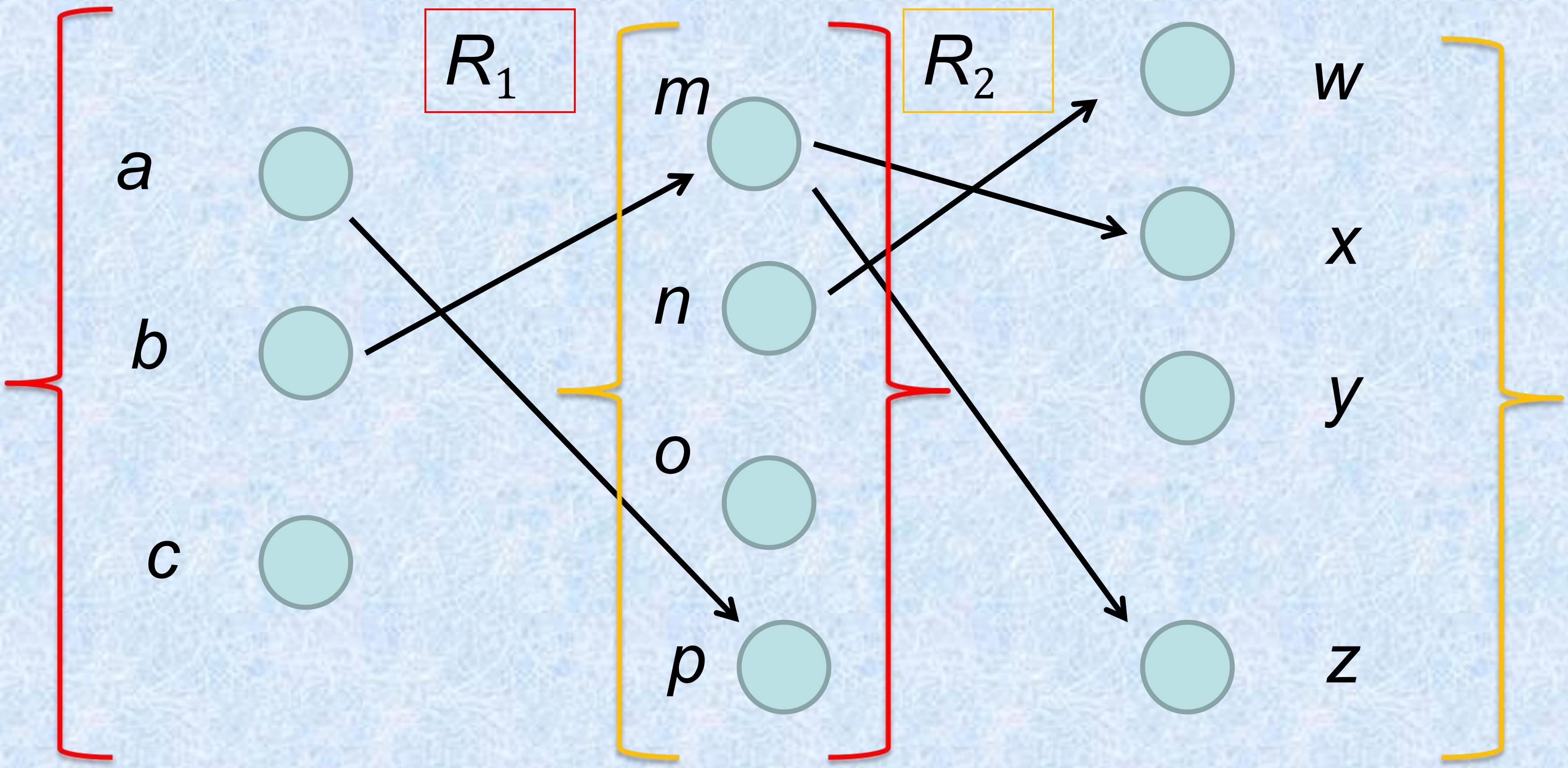
$$R1 \oplus R2$$

19: R1 “less than” R2 “greater than”

Composition

- Let R be a relation from a set A to a set B and S a relation from B to a set C. The composite of R and S is the relation consisting of ordered pairs (a, c), where $a \in A$, $c \in C$, and for which there exists an element $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$. We denote the composite of R and S by $S \circ R$.

Representing the Composition of a Relation



$R_2 \circ R_1 = \{(b,x),(b,z)\}$

Example 20

- What is the composite of the relations R and S, where R is the relation from {1,2,3} to {1,2,3,4} with R= {(1,1),(1,4),(2,3),(3,1),(3,4)} and S is the relation from {1,2,3,4} to {0,1,2} with S={(1,0),(2,0),(3,1),(3,2),(4,1)}?

Powers of a Relation

Definition: Let R be a binary relation on A. Then the powers R^n of the relation R can be defined inductively by:

- Basis Step: $R^1 = R$
- Inductive Step: $R^{n+1} = R^n \circ R$

(*see the slides for Section 9.3 for further insights*)

The powers of a transitive relation are subsets of the relation. This is established by the following theorem:

Theorem 1: The relation R on a set A is transitive iff $R^n \subseteq R$ for $n = 1,2,3 \ldots$.

(*see the text for a proof via mathematical induction*)

Powers of relations

- The *n*th power of a relation R for $n \geq 1$ is defined recursively as
  - $R^1 = R$
  - $R^{n+1} = R^n \circ R$

Example 22

- Let $R=\{(1,1),(2,1),(3,2),(4,3)\}$.Find the power R^n,$n=2,3,4,\ldots$

Theorem 1

- The relation R on a set A is transitive if and only if

$$R^n \subseteq R$$

- for n=1,2,3,…

§ 9.2: *n*-ary Relations

- An *n*-ary relation R on sets $A_1,\ldots,A_n$, written (with signature) $R: A_1 \times \ldots \times A_n$ or $R: A_1,\ldots,A_n$, is simply a subset
$$R \subseteq A_1 \times \ \ldots \ \times A_n.$$
- The sets A_i are called the *domains* of R.
- The *degree* of R is n.
- R is *functional in the domain A_i* if it contains at most one *n*-tuple $(\ldots, a_i ,\ldots)$ for any value a_i within domain A_i.

Relational Databases

- A *relational database* is essentially just an *n*-ary relation R.
- A domain A_i is a *primary key* for the database if the relation R is functional in A_i.
- A *composite key* for the database is a set of domains $\{A_i, A_j, \ldots\}$ such that R contains at most 1 *n*-tuple $(\ldots,a_i,\ldots,a_j,\ldots)$ for each composite value $(a_i, a_j,\ldots) \in A_i \times A_j \times \ldots$

Relational databases

- A model for databases, based on relations
- A database is viewed as a collection of *records*, each being a tuple of the form ($value_1$, $value_2$, ..., $value_n$)
- Because this relation is often given as a table, this term (*table*) is used as an equivalent for relation in database terminology

Database example

Student name	*ID*	*Major*	*GPA*
John	1022	Computer Science	3.07
Ellen	2773	Mathematics	3.65
Mark	4975	Biology	2.88

Database intension and extension

- The *fields* of the database (the columns) correspond to the domains of the relation
- The number of fields is the degree *n*
- The *intension* of the database (or its *schema*) is the permanent part of its design, i.e., the fields and the constraints on them
- The *extension* of the database is its current population of records

Database keys

- In order to uniquely identify tuples in a database, we need to know where to look, that is specify a field that serves for this purpose. This is the *primary key*.
- Care must be taken to maintain the key property when inserting records
- Some databases may have no primary key; instead they use a *composite* key made up from multiple fields
  - A flight database: key = airline + flight number

Selection Operators

- Let A be any n-ary domain $A = A_1 \times \ldots \times A_n$, and let $C: A \rightarrow \{\mathbf{T}, \mathbf{F}\}$ be any *condition* (predicate) on elements (n-tuples) of A.
- Then, the *selection operator* s_C is the operator that maps any (n-ary) relation R on A to the n-ary relation of all n-tuples from R that satisfy C.
  - *I.e.*, $\forall R \subseteq A,\ s_C(R) = \{a \in R \mid s_C(a) = \mathbf{T}\}$

Selection Operator Example

- Suppose we have a domain
  A = StudentName $\times$ Standing $\times$ SocSecNos
- Suppose we define a certain condition on A,
  $UpperLevel(name, standing, ssn) :\equiv$
  $[(standing = \text{junior}) \vee (standing = \text{senior})]$
- Then, $s_{UpperLevel}$ is the selection operator that takes any relation R on A (database of students) and produces a relation consisting of *just* the upper-level classes (juniors and seniors).

Projection Operators

- Let $A = A_1 \times \ldots \times A_n$ be any *n*-ary domain, and let $\{i_k\}=(i_1,\ldots,i_m)$ be a sequence of indices all falling in the range 1 to *n*,
  - That is, where $1 \leq i_k \leq n$ for all $1 \leq k \leq m$.
- Then the *projection operator* on *n*-tuples

  is defined by:

$$P_{\{i_k\}} : A \rightarrow A_{i_1} \times \ldots \times A_{i_m}$$

$$P_{\{i_k\}}(a_1,\ldots,a_n) = (a_{i_1},\ldots,a_{i_m})$$

Projection Example

- Suppose we have a ternary (3-ary) domain *Cars*=*Model*$\times$ *Year*$\times$ *Color*. (note n=3).
- Consider the index sequence $\{i_k\}$= 1,3. (m=2)
- Then the projection $P_{\{i_k\}}$ simply maps each tuple (a_1,a_2,a_3) = (*model*,*year*,*color*) to its image:

$$(a_{i_1}, a_{i_2}) = (a_1, a_3) = (model, color)$$

- This operator can be usefully applied to a whole relation $R\subseteq Cars$ (a database of cars) to obtain a list of the model/color combinations available.

Join Operator

- Puts two relations together to form a sort of combined relation.
- If the tuple (A,B) appears in R_1, and the tuple (B,C) appears in R_2, then the tuple (A,B,C) appears in the join $J(R_1,R_2)$.
  - A, B, and C here can also be sequences of elements (across multiple fields), not just single elements.

Join Example

- Suppose R_1 is a teaching assignment table, relating *Professors* to *Courses*.
- Suppose R_2 is a room assignment table relating *Courses* to *Rooms*,*Times*.
- Then $J(R_1,R_2)$ is like your class schedule, listing (*professor*,*course*,*room*,*time*).

Association Rules from Data Mining

- The count of an itemset I, denoted by σ (I), in a set of transactions $T = \{t_1, t_2, \ldots, t_k\}$, where k is a positive integer, is the number of transactions that contain this itemset. That is, $\sigma(I) = |\{t_i \in T \mid I \subseteq t_i\}|$.
- The support of an itemset I is the probability that I is included in a randomly selected transaction from T. That is, $support(I) = \sigma(I)/|T|$.

§ 9.3: Representing Relations

- Some ways to represent *n*-ary relations:
  - With an explicit list or table of its tuples.
  - With a function from the domain to **{T,F}**.
    - Or with an algorithm for computing this function.
- Some special ways to represent binary relations:
  - With a zero-one matrix.
  - With a directed graph.

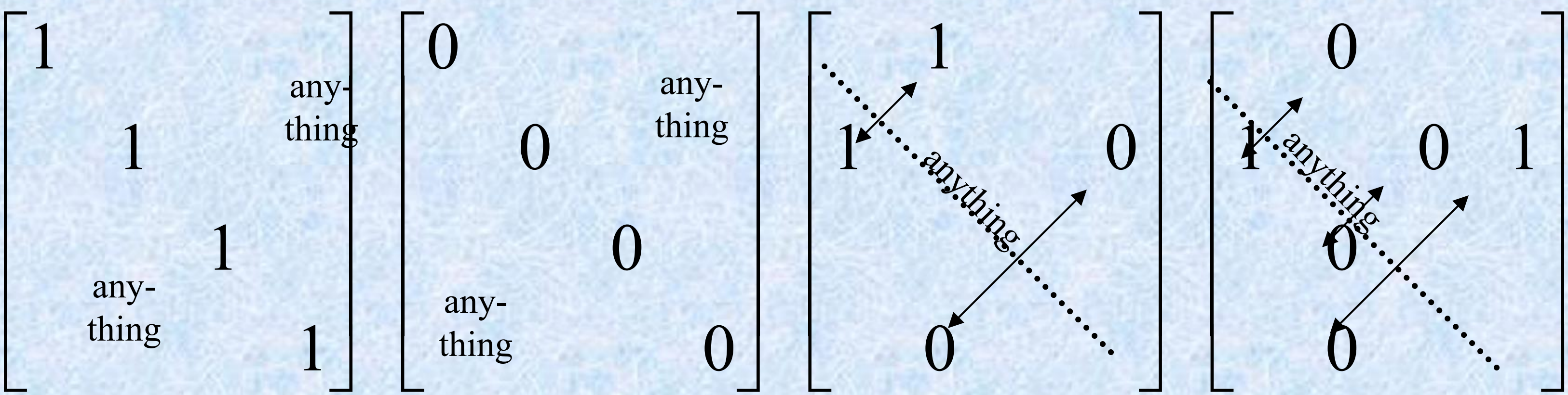
Using Zero-One Matrices

- To represent a binary relation $R:A\times B$ by an $|A|\times|B|$ 0-1 matrix $\mathbf{M}_R = [m_{ij}]$, let $m_{ij} = 1$ iff $(a_i,b_j)\in R$.
- *E.g.*, Suppose Joe likes Susan and Mary, Fred likes Mary, and Mark likes Sally.
- Then the 0-1 matrix representation of the relation Likes:Boys$\times$Girls relation is:

$$\begin{array}{r c} & \begin{array}{ccc} \text{Susan} & \text{Mary} & \text{Sally} \end{array} \\ \begin{array}{r} \text{Joe} \\ \text{Fred} \\ \text{Mark} \end{array} & \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{array}$$

Zero-One Reflexive, Symmetric

- Terms: *Reflexive*, *non-reflexive, irreflexive*, *symmetric, asymmetric, and antisymmetric*.
  - These relation characteristics are very easy to recognize by inspection of the zero-one matrix.



Reflexive: all 1's on diagonal

Irreflexive: all 0's on diagonal

Symmetric: all identical across diagonal

Antisymmetric: all 1's are across from 0's

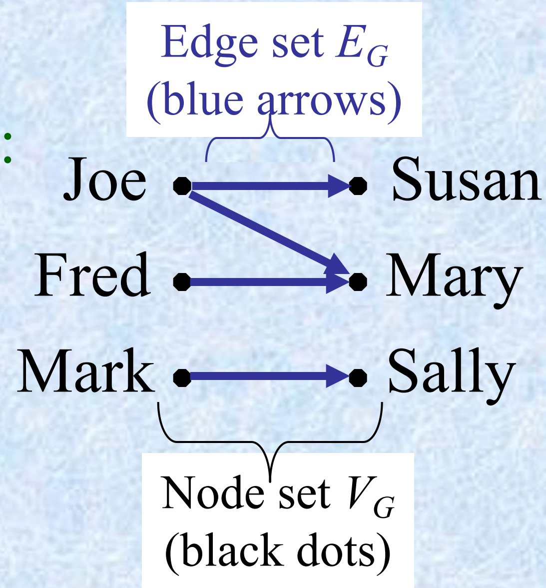
Using Directed Graphs

- A *directed graph* or *digraph* $G=(V_G,E_G)$ is a set V_G of *vertices (nodes)* with a set $E_G \subseteq V_G \times V_G$ of *edges (arcs,links)*. Visually represented using dots for nodes, and arrows for edges. Notice that a relation $R{:}A \times B$ can be represented as a graph $G_R=(V_G=A \cup B, E_G=R)$.

Matrix representation $\mathbf{M}_R$:

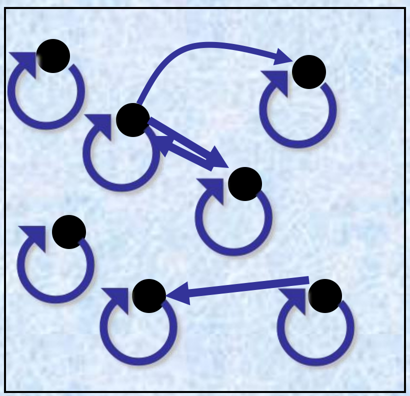
$$
\begin{array}{r}
 \\ \text{Joe} \\ \text{Fred} \\ \text{Mark}
\end{array}
\begin{array}{c}
\begin{array}{ccc} \text{Susan} & \text{Mary} & \text{Sally} \end{array} \\
\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}
\end{array}
$$

Graph rep. G_R:

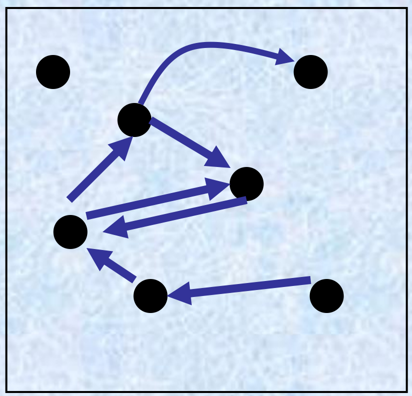


Digraph Reflexive, Symmetric

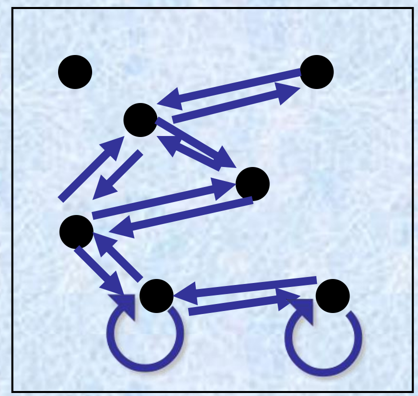
It is extremely easy to recognize the reflexive/irreflexive/ symmetric/antisymmetric properties by graph inspection.



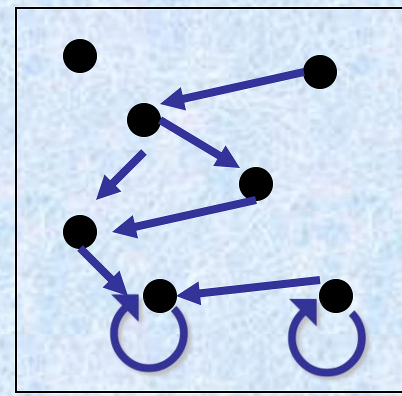
Reflexive:
Every node
has a self-loop



Irreflexive:
No node
links to itself



Symmetric:
Every link is
bidirectional



Antisymmetric:
No link is
bidirectional

These are asymmetric & non-antisymmetric

These are non-reflexive & non-irreflexive

关系特征、关系图、关系矩阵

关系特征	关系图特征	关系矩阵特征
自反	每一结点处有一环	对角线元素均为1
反自反	每一结点处有无环	对角线元素均为0
对称	两结点间有相反的两边同时出现	矩阵为对称矩阵
反对称	两结点间没有相反的边成对出现	当分量$C_{ij}=1(i\neq j)$时$C_{ji}=0$
传递	如果结点v_1,v_2间有边，v_2,v_3间有边，则v_1,v_3间必有边。	$R^n \subseteq R$

Example

$$M_{R_1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad M_{R_2} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad M_{R_2}^2 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$M_{R_3} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad M_{R_4} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad M_{R_4}^2 = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$M_{R_5} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad M_{R_6} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad M_{R_6}^2 = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

二元关系的运算

Relation Operation and Closures

二元关系

作为集合，存在并，交，差，非和对称差的运算；

作为函数 也存在复合运算和逆运算；

作为特殊的集合和函数的推广 还存在闭包运算。

作业

- § 9.1 – 42, 49, 50
- § 9.2 – 8, 20
- § 9.3 – 14, 32
- § 9.4 – 20, 22, 28

§ 9.4: Closures of Relations

- For any property *X*, the "*X* closure" of a set *A* is defined as the "smallest" superset of *A* that has the given property.

关系 R 的 P 闭包就是包含 R 且具有 P 中一切性质的那些关系中的最小的那个.

§ 9.4: Closures of Relations

- The *reflexive closure* of a relation R on A is obtained by adding (a,a) to R for each $a \in A$. *I.e.*, it is $\boxed{R \cup I_A}$
- The *symmetric closure* of R is obtained by adding (b,a) to R for each (a,b) in R. *I.e.*, it is $\boxed{R \cup R^{-1}}$
- The *transitive closure* or *connectivity relation* of R is obtained by repeatedly adding (a,c) to R for each $(a,b),(b,c)$ in R.
  - *I.e.*, it is

$$\boxed{R^* = \bigcup_{n \in \mathbf{Z}^+} R^n}$$

关系的闭包/Closure of Relations

[定义1]：称 $R \cup I_A$ 是 R 的自反闭包/reflexive closure of R，记为r(R)。

[定理1]：设 R 是 A 上的二元关系，则r(R)是包含 R，具有自反性的最小关系。

证明：对 $\forall x \in A$，$(x, x) \in I_A \subseteq R \cup I_A$，且 $R \subseteq R \cup I_A$。

若 R' 为包含 R 且具有自反性，则 $I_A \subseteq R'$，$R \subseteq R'$，$I_A \cup R \subseteq R'$。即 $I_A \cup R$ 为最小。

[推论1]：R是自反闭包当且仅当R是自反的。

证明：

（1）R是自反闭包，$R = I_A \cup R \Rightarrow I_A \subseteq R$；

（2）R具有自反性，$I_A \subseteq R$，$R \cup I_A = R$；

[定义2]：称 $R \cup R^{-1}$ 是 R 的对称闭包/symmetric closure of R，记为 s(R)。

[定理2]：设 R 是 A 上的二元关系，则 $R \cup R^{-1}$ 是对称的，包含 R 的最小关系，

（其中 R^{C} 是 R 的逆关系）。

证明：若 $(a, b) \in R \cup R^{-1}$，则

$(a, b) \in R$ 或 $(a, b) \in R^{-1}$，

$\Rightarrow (b, a) \in R^{-1}$ 或 $(b, a) \in R$

故 $(b, a) \in R \cup R^{-1}$（对称性），

若 R’ 为包含 R，对称的二元关系，

则对 $\forall (a, b) \in R \cup R^{-1}$，

若 $(a, b) \in R \Rightarrow (a, b) \in R'$；

若 $(a, b) \in R^{-1} \Rightarrow (b, a) \in R$，

$\Rightarrow (b, a) \in R'$

又 R’ 具有对称性，$(a, b) \in R'$，

故 $R \cup R^{-1} \subseteq R'$ 。

[推论2]：当且仅当R是对称闭包时，R具有对称性。

证明：（1）R具有对称性，

若（a，b）$\in$R，则（b，a）$\in$R，

又（b，a）$\in R^{-1}$

即$\forall$（b，a）$\in R^{-1}$，（a，b）$\in$R

$\Rightarrow$（b，a）$\in$R

$\Rightarrow R^{-1} \subseteq R \Rightarrow R \cup R^{-1} = R$；

（2）R是对称闭包时，$R = R \cup R^{-1} \Rightarrow$ R具有对称性。

Examples

- If $A = Z$, then $r(\neq) =$
- $Z \times Z$
- If $A = Z^+$, then $s(<) =$
- $\neq$.
- What is the (infinite) connection matrix of $s(<)$?
- If $A = Z$, then $s(\leq) =$?

Paths in Digraphs/Binary Relations

- A *path* of length n from node a to b in the directed graph G (or the binary relation R) is a sequence $(a,x_1), (x_1,x_2), \ldots, (x_{n-1},b)$ of n ordered pairs in E_G (or R).
  - An empty sequence of edges is considered a path of length 0 from a to a.
  - If any path from a to b exists, then we say that a is *connected to b*. ("You can get there from here.")
- A path of length $n \geq 1$ from a to itself is called a *circuit* or a cycle.
- Note that there exists a path of length n from a to b in R if and only if $(a,b) \in R^n$.

Theorem 1

- Let *R* be a relation on a set *A*. then R^∞ is the transitive closure of *R*.

$$t(R) = R^\infty = R \cup R^2 \cup R^3 \cup \cdots = \bigcup_{i=1}^{\infty} R^i$$

- Proof: we must show that R^∞
  - 1) is a transitive relation
  - 2) contains *R*
  - 3) is the smallest transitive relation which contains *R*

Proof of Part 1)

- Suppose (x, y) and (y, z) are in R^∞. Show (x, z) is in R^∞.
  - By definition of R^∞, (x, y) is in R^m for some m and (y, z) is in R^n for some n.
  - Then (x, z) is in $R^n \circ R^m = R^{m+n}$ which is contained in R^∞.
  - Hence, R^∞ must be transitive.

Proof of Part 2)

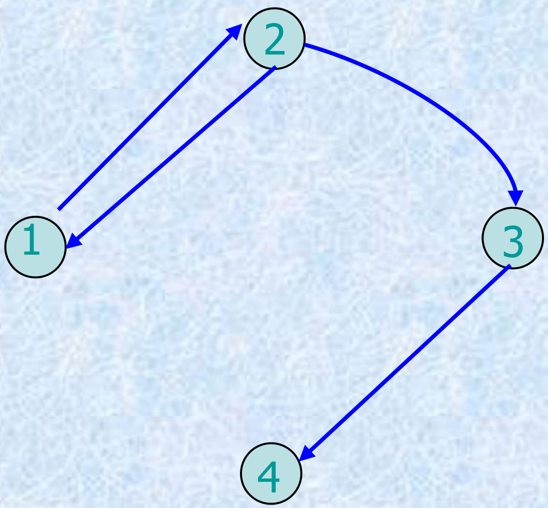
- Easy from the definition of R^{∞}

Proof of Part 3)

- Now suppose S is any transitive relation that contains R, show S contains R^∞ (that is R^∞ is the smallest such relation).
- $R \subseteq S$ so $R^2 \subseteq S^2 \subseteq S$ since S is transitive
- Therefore $R^n \subseteq S^n \subseteq S$ for all n. (why?)
- Hence S must contain R^∞ since it must also contain the union of all the powers of R.
  - Q. E. D.
  - In fact, we need only consider paths of length n or less.

Example 1

- Let
  - A={1, 2, 3, 4}
  - R={(1, 2), (2, 3), (3, 4), (2, 1)}
- Find the transitive closure of R.



$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 1

$$\left(M_R\right)_{\odot}^{2} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \left(M_R\right)_{\odot}^{4} = \left(M_R\right)_{\odot}^{6} = \ldots$$

$$\left(M_R\right)_{\odot}^{3} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \left(M_R\right)_{\odot}^{5} = \left(M_R\right)_{\odot}^{7} = \ldots$$

$$M_{R^{\infty}} = M_R \vee \left(M_R\right)_{\odot}^{2} \vee \left(M_R\right)_{\odot}^{3} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Theorem 2

- Let A be a set with $|A|=n$, and let R be a relation on A. Then

$$R^{\infty} = \bigcup_{i=1}^{n} R^{i} = R \cup R^{2} \cup \cdots \cup R^{n}$$

Proof

$$R^{\infty} = \bigcup_{i=1}^{\infty} R^i = R \cup R^2 \cup R^3 \cup \cdots$$

$$? = \bigcup_{i=1}^{n} R^i = R \cup R^2 \cup \cdots \cup R^n$$

- The equality will hold, if, for $k \le n < m$, we have
  - $R^m \subseteq R^k$
  - $(a, b) \in R^m \to (a, b) \in R^k$

Proof

- Let a and b be A and suppose that a, x_1, x_2, …, x_{m-1}, b is a path of length m from a to b in R
  - $(a, x_1) \in R$
  - $(x_1, x_2) \in R$
  - …
  - $(x_{m-1}, b) \in R$

Proof

- There are $m+1$ elements in the path, but we have only n distinct elements in A.
  - So, there must be some same vertex in the path, say $x_i = x_j = c, i<j$
    - $(a, x_1) \in R$
    - $(x_1, x_2) \in R$
    - …
    - $(x_{i-1}, x_i) \in R$
    - $(x_i, x_{i+!}) \in R$
    - …
    - $(x_{j-1}, x_j) \in R$
    - $(x_j, x_{j+1}) \in R$
    - …
    - $(x_{m-1}, b) \in R$
- The red edges form a cycle in the path, we get a new path by deleting the cycle

Proof

- A new path from *a* to *b* by deleting the cycle
  - $(a, x_1) \in R$
  - $(x_1, x_2) \in R$
  - …
  - $(x_{i-1}, x_i) \in R$
  - $(x_j, x_{j+1}) \in R$
  - …
  - $(x_{m-1}, b) \in R$

Proof

- A path from a to b ($x_i = x_j = c$)
  - $a, x_1, x_2, \ldots, x_{i-1}, c, x_{j+1}, \ldots, x_{m-1}, b$
- The length is $k = m - j + i$.
- The process can continue until $k \le n$, so we have
  - $R^m \subseteq R^k$
  - $\forall m\ (m > n \wedge (a, b) \in R^m \rightarrow \exists k\ (k \le n \wedge (a, b) \in R^k))$
- Therefore

$$R^\infty = \bigcup_{i=1}^{n} R^i = R \cup R^2 \cup \cdots \cup R^n$$

– QED

Simple Transitive Closure Alg.

A procedure to compute R^* with 0-1 matrices.

procedure *transClosure*($\mathbf{M}_R$:rank-n 0-1 mat.)

$\mathbf{A} := \mathbf{B} := \mathbf{M}_R$;

for $i := 2$ to n **begin**

$\mathbf{A} := \mathbf{A} \odot \mathbf{M}_R$; $\mathbf{B} := \mathbf{B} \vee \mathbf{A}$ {join}

end {note **A** represents R^i, **B** represents $\bigcup_{j=1}^{i} R^i$ }

return B {Alg. takes $\Theta(n^4)$ time}

Some definitions

- Let
  - $A = \{a_1, a_2, \ldots, a_n\}$
  - R be a relation on A
- *Interior vertices*
  - $a, x_1, x_2, \ldots, x_i, b$

Some definitions

- W_k: a Boolean matrix, for $1 \leq k \leq n$
  - W_k has a 1 in position i, j
    - If and only if
  - there is a path from a_i to a_j in R whose interior vertices, if any, come from the set $\{a_1, a_2, \ldots, a_k\}$
- What about W_0 W_n ?
  - Let $W_0 = W_R$
  - $W_n = W^{\infty}$
  - $W_0, W_1, W_2, \ldots, W_n$

Warshall's Algorithm

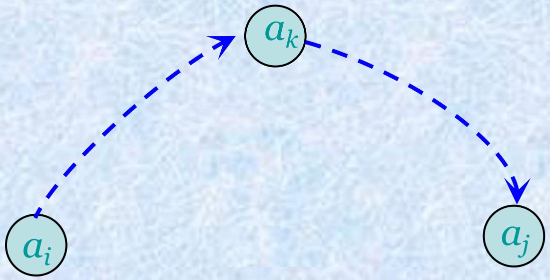
- Procedure
  - begin with the matrix of R, and
  - compute each matrix , W_{k-1} from the previous matrix W_k, and,
  - reach W^{∞} in n steps,

Warshall's Algorithm

- Suppose
  - $W_k = [t_{ij}]$
  - $W_{k\text{-}1} = [s_{ij}]$
- If $t_{ij} = 1$, then there must be a path from a_i to a_j whose interior vertices come from the set $\{a_1, a_2, \ldots, a_k\}$.
  - Whether a_k is an interior vertex ?
  - Two cases

Warshall's Algorithm

- a_k is not an interior vertex
  - then all interior vertices must actually come from the set $\{a_1, a_2, \ldots, a_{k-1}\}$
  - so $s_{ij} = 1$.
- a_k is an interior vertex
  - Assume a_k appears only once (why?)
  - Two subpaths
    - a_i to a_k and a_k to a_j
  - $s_{ik} = 1$ and $s_{kj} = 1$



Warshall's Algorithm

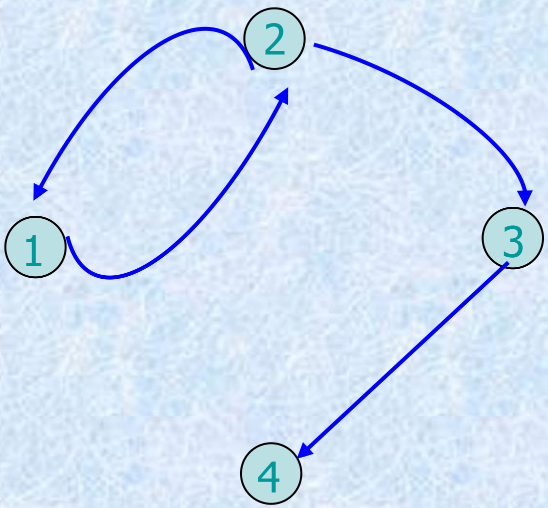
- The basis for Warshall's Algorithm
  - $t_{ij} = 1$
    - If and only if
  - either
    - $s_{ij} = 1$
    - $s_{ik} = 1$ and $s_{kj} = 1$

Warshall's Algorithm

- *Step1*:
  – First transfer to W_k all 1's in $W_{k\text{-}1}$.
- *Step2*:
  – List the locations p_1, p_2, …, in column k of $W_{k\text{-}1}$, where the entry is 1
  – List the locations q_1, q_2, …, in row k of $W_{k\text{-}1}$, where the entry is 1
- *Step3*:
  – Put 1's in all the positions p_i, q_j of W_k (if they are not already there)

Example 2 (1)

- Let
  - A={1, 2, 3, 4}
  - R={(1, 2), (2, 3), (3, 4), (2, 1)}
- Find the transitive closure of R.



$$M_R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 2

$$W_0 = \begin{bmatrix} \underline{0} & \underline{\underline{1}} & \underline{0} & \underline{0} \\ \underline{\underline{1}} & 0 & 1 & 0 \\ \underline{0} & 0 & 0 & 1 \\ \underline{0} & 0 & 0 & 0 \end{bmatrix} = M_R, \quad W_1 = \begin{bmatrix} 0 & \underline{\underline{1}} & 0 & 0 \\ \underline{\underline{1}} & \underline{\underline{1}} & \underline{\underline{1}} & \underline{0} \\ 0 & \underline{0} & 0 & 1 \\ 0 & \underline{0} & 0 & 0 \end{bmatrix}$$

$$W_2 = \begin{bmatrix} 1 & 1 & \underline{\underline{1}} & 0 \\ 1 & 1 & \underline{\underline{1}} & 0 \\ \underline{0} & \underline{0} & \underline{0} & \underline{\underline{1}} \\ 0 & 0 & \underline{0} & 0 \end{bmatrix}, \quad W_3 = \begin{bmatrix} 1 & 1 & 1 & \underline{\underline{1}} \\ 1 & 1 & 1 & \underline{\underline{1}} \\ 0 & 0 & 0 & \underline{\underline{1}} \\ \underline{0} & \underline{0} & \underline{0} & \underline{0} \end{bmatrix} = W_4 = W^{\infty}$$

Warshall’s Algorithm

```
Algorithm Warshall
CLOSURE←MAT
   For k = 1 thru N
     For i = 1 thru N
       For j = 1 thru N
         CLOSURE[i,j]←CLOSURE[i,j]
                    ∨(CLOSURE[i,k]∧CLOSURE[k,j])
End of Algorithm Warshall
```

Analysis

- Complexity of Algorithm
  - Warshall
    - n^3
  - M^{∞}
    - n^4 $= M_R \vee \left(M_R\right)_{\odot}^{2} \vee \ldots \vee \left(M_R\right)_{\odot}^{n}$

$Warshall$ 算法

1) 置新矩阵 $A := M$

2) $i := 1$

3) 对所有 j,若 $A[j,i]=1$,则对

$k=1,2,\cdots,n$

$A[j,k]:=A[j,k]+A[i,k]$

j行 $\leftarrow j$行 $+\,i$行

4) $i+1$

5) 若$i\leq n$,则转3),否则停止.

例 6：

设A＝｛1， 2， 3｝， R＝｛（1， 1），（1， 2），(2， 3）｝求R* 。

解: $$R=\begin{bmatrix}1&1&0\\0&0&1\\0&0&0\end{bmatrix},\ R^2=\begin{bmatrix}1&1&1\\0&0&0\\0&0&0\end{bmatrix}=R^3\ \square$$

故 $$R^*=R\cup R^2\cup R^3=\begin{bmatrix}1&1&1\\0&0&1\\0&0&0\end{bmatrix}$$

$$R^*=\{(1,\ 1),(1,\ 2),(1,\ 3),(2,\ 3)\}$$

注： 当｜A｜＝n 较大时，用定理5计算 R^* 工作量非常大。

Warshall 算法：（1962）

设R是n个元素集合上的二元关系

(1) 是R的相关矩阵；

(2) $i \leftarrow 1$；

(3) for $j=1$ to n do

if $a_{j,i}=1$ then do

for $k=1$ to n do

$a_{j,k} \leftarrow a_{j,k} \vee a_{i,k}$

(4) $i=i+1$；

(5) if $i \le n$， then **go (3)**

else stop

$$R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

§ 9.5: Equivalence Relations

- An *equivalence relation* (e.r.) on a set A is simply any binary relation on A that is reflexive, symmetric, and transitive.
  - *E.g.*, **=** itself is an equivalence relation.
  - For any function f:$A \rightarrow B$, the relation "have the same f value", or $=_f :\equiv \{(a_1,a_2) \mid f(a_1)=f(a_2)\}$ is an equivalence relation,
    - *e.g.*, let m="mother of" then $=_m$ = "have the same mother" is an e.r.

Equivalence Relations

[定义1] 等价关系:

A上的二元关系R，如果R是

(1）自反的

(2）对称的

(3) 传递的

称R为等价关系/Equivalence Relations 。

（a， b）$\in$R，称a与b等价。

Equivalence Relation Examples

- "Strings *a* and *b* are the same length."
- "Integers *a* and *b* have the same absolute value."
- "Real numbers *a* and *b* have the same fractional part." (*i.e.*, $a - b \in \mathbf{Z}$)
- "Integers *a* and *b* have the same residue modulo *m*." (for a given $m>1$)

Equivalence Classes

- Let R be any equiv. rel. on a set A.
- The *equivalence class* of a,

  $[a]_R :\equiv \{ b \mid aRb \}$ (optional subscript R)

  – It is the set of all elements of A that are "equivalent" to a according to the eq.rel. R.

  – Each such b (including a itself) is called a *representative* of $[a]_R$.
- Since $f(a)=[a]_R$ is a function of a, any equivalence relation R can be defined using $aRb :\equiv$ "a and b have the same f value", given f.

Equivalence Class Examples

- "Strings *a* and *b* are the same length."
  - [*a*] = the set of all strings of the same length as *a.*
- "Integers *a* and *b* have the same absolute value."
  - [*a*] = the set $\{a, -a\}$
- "Real numbers *a* and *b* have the same fractional part (*i.e.*, $a - b \in \mathbf{Z}$)."
  - [*a*] = the set $\{\ldots, a-2, a-1, a, a+1, a+2, \ldots\}$
- "Integers *a* and *b* have the same residue modulo *m*." (for a given $m>1$)
  - [*a*] = the set $\{\ldots, a-2m, a-m, a, a+m, a+2m, \ldots\}$

Example

- Let $A = Z$ and let
  - $R = \{(a, b) \in A \times A \mid a$ and b yield the same remainder when divided by 2$\}$.
- Show
  - congruence mod 2 is an equivalence relation
- note
  - 2 is called the *modulus*
  - $a \equiv b \pmod{2}$
    - “*a* is congruent to *b* mod 2.“(模2同余)

Solution, *a typical way*

- First
  - clearly $a \equiv a \pmod{2}$.
  - Thus R is *reflexive*.
- Second
  - if $a \equiv b \pmod{2}$, then a and b yield the same remainder when divided by 2, so $b \equiv a \pmod{2}$.
  - R is *symmetric*.
- Finally
  - suppose that $a \equiv b \pmod{2}$ and $b \equiv c \pmod{2}$. Then a, b, and c yield the same remainder when divided by 2.
  - Thus, $a \equiv c \pmod{2}$.
  - R is *transitive*.
- Hence congruence mod 2 is an equivalence relation.

注：

等价关系R把A的元素分为若干类，各类之间没有公共元素。确定的R是对集合A进行的一个划分。

[**定义**3]集合的划分：把集合A分为若干子集A_1，A_2，…，满足：

（1）当$i \neq j$时$A_i \cap A_j = \Phi$

（2）$\forall a \in A$，$\exists i$，使$a \in A_i$（$i = 1, 2, \ldots$）

则集合$P_r(A) = \{A_1, A_2, \ldots, A_n, \ldots\}$称为A的一个划分/partition。

Partitions

- A *partition* of a set A is the set of all the equivalence classes $\{A_1, A_2, \dots\}$ for some equivalence relation on A.
  - The A_i's are all disjoint and their union = A.
- They "partition" the set into pieces. Within each piece, all members of that set are equivalent to each other.

Equivalence Relations and Partitions

- If ***P*** is a partition of a set *A*, then ***P*** can be used to construct an equivalence relation on *A*.

Theorem 1

- Let ***P*** be a partition of a set *A*. Define the relation *R* on *A* as follows:
  - *a R b*
    - If and only if
  - *a* and *b* are members of the same block.
- Then *R* is an equivalence relation on *A*.

Proof

  - If $a \in A$, then clearly a is in the same block as itself; so a R a.
  - If $a\ R\ b$, then a and b are in the same block; so $b\ R\ a$.
  - If $a\ R\ b$ and $b\ R\ c$, then a, b, and c must all lie in the same block of ***P***. Thus $a\ R\ c$.
- Since *R* is reflexive, symmetric, and transitive, R is an equivalence relation.
- *R* will be called the *equivalence relation determined by* ***P***.

– QED

Example

- Let A = {1, 2, 3, 4} and consider the partition $\boldsymbol{P}$ = {{1, 2, 3}, {4}} of A. Find the equivalence relation R on A determined by $\boldsymbol{P}$.
- Solution
  - The blocks of $\boldsymbol{P}$ are {l, 2, 3} and {4}.
  - Each element in a block is related to every other element in the same block and only to those elements.
  - Thus R = {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, l), (3, 2), (3, 3), (4, 4)}.

Lemma 1(引理)

- Let R be an equivalence relation on a set A, and let $a \in A$ and $b \in A$.
- Then
  - $a \; R \; b$
    - if and only if
  - $R(a) = R(b)$

Proof of lemma 1

- First suppose that $R(a) = R(b)$
  - Since R is reflexive, $b \in R(b) = R(a)$, so $a\ R\ b$.
- Conversely, suppose that $a\ R\ b$, show that $R(a) = R(b)$
  - choose $x \in R(b)$
  - bRx
  - $a\ R\ b$
  - $a\ Rx$ (transitive)
  - $x \in R(a)$.
  - Thus $R(b) \subseteq R(a)$
  - A completely similar argument shows that, $R(a) \subseteq R(b)$
- so $R(a) = R(b)$

  – QED

Theorem 2

- Let R be an equivalence relation on A, and let ***P*** be the collection of all distinct relative sets $R(a)$ for a in A.
- Then ***P*** is a partition of A, and R is the equivalence relation determined by ***P***.

- R = {(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, l), (3, 2), (3, 3), (4, 4)}
- R(1)=R(2)=R(3)={1,2,3} ,R(4)={4}
- ***P*** ={R(1),R(4)}

Proof

- P = $\{R(a) \mid a \in A\}$
  - Every element of A belongs to some relative set.
    - Since reflexivity of R, $a \in R(a)$
  - If $R(a)$ and $R(b)$ are not identical, then $R(a) \cap R(b) = \varnothing$
    - If $R(a) \cap R(b) \neq \varnothing$, then $R(a) = R(b)$.
    - Contrapositive(换质位命题、对换式、逆反式)

Proof :

If $R(a) \cap R(b) \neq \emptyset$, then $R(a) = R(b)$

- Let $c \in R(a) \cap R(b)$
  - $a\ R\ c$, $b\ R\ c$
  - $c\ R\ b$, since R is symmetric
  - $a\ R\ b$, since R is transitive
  - $R(a) = R(b)$ by Lemma I
- So, ***P*** is a partition
- $a\ R\ b$ iff a and b belong to the same block of ***P***, Thus ***P*** determines R.

– QED

Equivalence classes

- If R is an equivalence relation on A, then the sets $R(a)$ are traditionally called *equivalence classes* of R, denoted by $[a]$
- The partition ***P*** consists of all equivalence classes of R is denoted by A/R and called
  - the *quotient set*, or
  - *the partition of A induced by R,* or,
  - *A modulo R*.

Example 7

- Let
  - $A = \{1, 2, 3, 4\}$
  - $R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 3), (3, 3), (4, 4)\}$.
- Determine A/R.
- Solution
  - $R(1) = \{1, 2\} = R(2)$
  - $R(3) = \{3, 4\} = R(4)$.
  - Hence $A/R = \{\{1, 2\}, \{3, 4\}\}$

Determining partitions A/ R

- *STEP 1*: Choose any element of A and compute the equivalence class $R(a)$.
- *STEP 2*: if $R(a) \neq A$, choose an element b, not included in $R(a)$, and compute the equivalence class $R(b)$.
- *STEP 3*: If A is not the union of previously computed equivalence classes, then choose an element x of A that is not in any of those equivalence classes and compute $R(x)$.
- *STEP 4*: Repeat step 3 until all elements of A are included in the computed equivalence classes. If A is countable, this process could continue indefinitely. In that case, continue until a pattern emerges that allows you to describe or give a formula for all equivalence classes.

Example 8

- Let *R* be the equivalence relation defined in congruence modulo 2, determine *R/A*.
- Solution
  - $R(0)=\{\ldots, -6, -4, -2, 0, 2, 4, 6, \ldots\}$
  - $R(1)=\{\ldots, -5, -3, -1, 1, 3, 5, 7, \ldots\}$
  - Hence *A*/*R* consists of the set of even integers and the set of odd integers

Example 14

- What are the sets in the partition of the integers arising from congruence modulo 4?

例１：

$A=\{52$张扑克$\}$

$R_1=\{(a, b) \mid a$与b同花，a, b是扑克$\}$

$R_2=\{(a, b) \mid a$与b同点，a, b是扑克$\}$

则：R_1把A分为四类同花类，

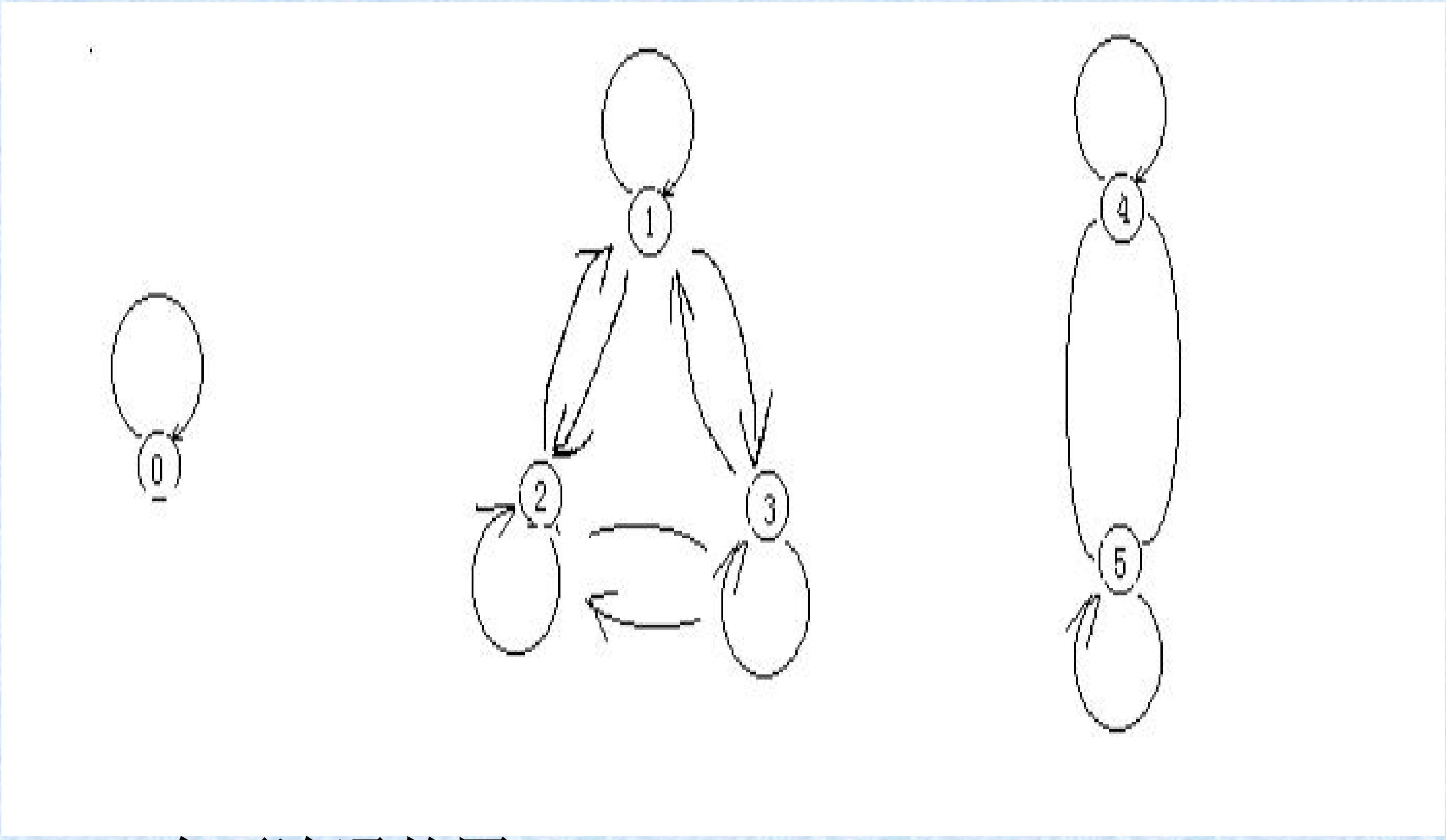
R_2把A分为13类同点类。

例２：

$A=\{0, 1, 2, 3, 4, 5\}$

$R=\{(0, 0), (1, 1), (2, 2), (3, 3), (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (4, 4), (4, 5), (5, 4), (5, 5)\}$

R是等价关系，但不直观，用关系图表示。



三个不连通的图

二元关系R是自反的，对称的，传递的，且把A分成了三个等价类，（A）＝{{0}，{1，2，3}，{4，5}}

［定义］商集：等价关系R将A分成若干等价类，每个类选个代表组成新的集合称为A关于R的商集，表示为A/R。

例：在上例中

A/R＝{［0］，［1］，［4］}

[定义]：商集的元素个数（即A在R下的等价类个数）称为R的秩。

例3：

$R=\{(a,\ b)\mid a\equiv b\pmod 3,\ a,\ b\in I\}$ 是整数集合 I 上模 3 同余的二元关系.

证明R是等价关系。

证明：由于 $3\mid(a-b)$ $a\equiv b\pmod 3$

(1) $3\mid(a-a)$， 自反性

(2)若 $3\mid(a-b)$，则 $3\mid(b-a)$ 对称性

(3)若 $3\mid(a-b)$，$3\mid(b-c)$，则

$$a-c=(a-b)+(b-c)$$

$$\Rightarrow 3\mid(a-c)$$

满足传递性, 故R是等价关系

商集：$I/R=\{[0],\ [1],\ [2]\}$

其中 $[0]=\{\dots,\ -6,\ -3,\ 0,\ 3,\ 6,\ \dots\}$

$[1]=\{\dots,\ -5,\ -2,\ 1,\ 4,\ 7,\ \dots\}$

$[2]=\{\dots,\ -4,\ -1,\ 2,\ 5,\ 8,\ \dots\}$

例4：

N的一个分划为：若$10^{i-1} \leqslant a < 10^{i}$，则$a \in A_i$，对应的等价关系

$R = \{(a, b) \mid$ a与b有相同位数，$a, b \in N\}$ 相同位的数归为一类，此R把N分划为可列个等价类。

等价关系的运算

[定理2]：

若R_1，R_2是A上的等价关系，则$R_1 \cap R_2$也是等价关系。

证明：$R_1 \cap R_2 = R_3$，R_3传递。

(1) $\forall a \in A$，由R_1，R_2自反

$(a, a) \in R_1$，$(a, a) \in R_2$，于是

$(a, a) \in R_1 \cap R_2 = R_3$，$R_3$自反

(2) 若$(a, b) \in R_3$，则$(a, b) \in R_1$

且$(a, b) \in R_2$，由R_1，R_2对称

$(b, a) \in R_1$，$(b, a) \in R_2$

于是$(b, a) \in R_1 \cap R_2 = R_3$，$R_3$对称

(3)若（a， b）$\in R_3$，（b， c）$\in R_3$

$$\Rightarrow \begin{cases} (a,b)\in R_1, (b,c)\in R_1 \Rightarrow (a,c)\in R_1 \\ (a,b)\in R_2, (b,c)\in R_2 \Rightarrow (a,c)\in R_2 \end{cases}$$

于是（a， c）$\in R_1 \cap R_2 = R_3$， R_3传递。

归纳以上三点， $R_3 = R_1 \cap R_2$也是等价的二元关系。

[定理3]：若R_1，R_2是A上等价关系，则$R_1 \cup R_2$是相容关系。

注：$R_1 \cup R_2$不一定 是等价关系。

证明：

设 $R_4 = R_1 \cup R_2$

(1) $\forall a \in A$，$(a, a) \in R_1$，$(a, a) \in R_2$，于是 $(a, a) \in R_1 \cup R_2 = R_4$。

(2) 若 $(a, b) \in R_4$，$(a, b) \in R_1$ 或 $(a, b) \in R_2$ 由 R_1，R_2 对称性，$(b, a) \in R_1$ 或 $(b, a) \in R_2$，

于是 $(b, a) \in R_1 \cup R_2 = R_4$

注：$R_4 = R_1 \cup R_2$ 是相容的，即

$R_1 \cup R_2$ 不一定是等价关系

若 $(a, b) \in R_4$，$(b, c) \in R_4$

(a, b) 与 (b, c) 可能分别属于

R_1 与 R_2，无法保证 $R_1 \cup R_2$ 的传递性。

例 5：

$A=\{a, b, c\}$

$R_1=\{(a, a), (b, b),$
$(c, c), (a, b), (b, a)\}$

$R_2=\{(a, a), (b, b),$
$(c, c), (b, c), (c, b)\}$

$R_4=R_1\cup R_2=\{(a, a), (b, b),$
$(c, c), (a, b), (b, a),$
$(b, c), (c, b)\}$

显然，$(a, b), (b, c)\in R_1\cup R_2$

但$(a, c)\notin R_1\cup R_2$（不具有传递性），

但$(R_1\cup R_2)^*$一定是传递的。

[定理4]：若R_1，R_2是A上等价关系，

则（$R_1 \cup R_2$）*也是A上等价关系。

证明：

(1) $\forall a \in A$，$(a, a) \in (R_1 \cup R_2) \subseteq (R_1 \cup R_2)^*$

(2)（$R_1 \cup R_2$）是对称的，只需证明对称的二元关系的传递扩张还是对称的。

设R是A上对称的二元关系，R_1是R的传递扩张，若$(a, b) \in R_1$，则$(a, b) \in R$或$(a, b) \in R \cdot R$，

a. 若$(a, b) \in R$，$(b, a) \in R$，故$(b, a) \in R_1$;

b. 若$(a, b) \in R \cdot R$，则$\exists c$，使

$(a, c), (c, b) \in R$

$\Rightarrow (b, c), (c, a) \in R$

$\Rightarrow (b, a) \in R \cdot R$

$\Rightarrow (b, a) \in R_1$

由$(R_1 \cup R_2)$对称$\Rightarrow (R_1 \cup R_2)_1$也对称，…，

对任一k，$(R_1 \cup R_2)_k$也对称。

若$(a, b) \in (R_1 \cup R_2)^*$，则$\exists k$，

使$(a, b) \in (R_1 \cup R_2)_k$，

故$(b, a) \in (R_1 \cup R_2)_k \subseteq (R_1 \cup R_2)^*$，

于是$(R_1 \cup R_2)^*$是对称的。

(3) 任一传递闭包都具有传递性。

即证。

等价关系对应于集合的一种分划。

设R_1，R_2是A上等价关系，对应的分划为π_1，π_2，

$\pi_1 \cdot \pi_2$对应于$R_1 \cap R_2$，称为**二个分划的积，**使分划分更细；

$\pi_1 + \pi_2$对应于$(R_1 \cup R_2)^*$，称为两个**分划的和**，使分划比π_1，π_2要粗。

例6：

$A = \{a, b, c, d, e, f, g, h, i, j, k\}$

$$R_1 \leftrightarrow \pi_1 = \{ \underset{A_1}{\overline{abcd}}, \underset{A_2}{\overline{efg}}, \underset{A_3}{\overline{hi}}, \underset{A_4}{\overline{jk}} \}$$

$$R_2 \leftrightarrow \pi_2 = \{ \underset{B_1}{\overline{abch}}, \underset{B_2}{\overline{di}}, \underset{B_3}{\overline{efik}}, \underset{B_4}{\overline{g}} \}$$

R_1＝{（a， a），（b， b），（c， c），（d，d），（e， e），（f， f），（g， g），（h，h），（i， i），（j， j），（k， k），（a，b），（a， c），（a， d），（b， c），（b，d），（c， d），（b， a），（c， a），（d，a），（c， b），（d， b），（d， c），（e，f），（e， g），（f， g），（f， e），（g，e），（g， f），（h， i），（i， h），（j，k），（k， j）}

$$
\begin{aligned}
R_2 = \{ & (a, a), (b, b), (c, c), \\
& (d, d), (e, e), (f, f), \\
& (g, g), (h, h), (i, i), \\
& (j, j), (k, k), (a, b), \\
& (a, c), (a, h), (b, c), \\
& (b, h), (c, h), (b, a), \\
& (c, a), (h, a), (c, b), \\
& (h, b), (h, c), (d, i), \\
& (i, d), (e, f), (e, j), \\
& (e, k), (f, j), (f, k), \\
& (j, k), (f, e), (j, e), \\
& (k, e), (j, f), (k, f), \\
& (k, j) \}
\end{aligned}
$$

$R_1 \cap R_2 = \{ (a, a), (b, b), (c, c), (d, d), (e, e), (f, f), (g, g), (h, h), (i, i), (j, j), (k, k), (a, b), (a, c), (b, c), (e, f), (j, k), (b, a), (c, a), (c, b), (f, e), (k, j)$

$R_1 \cup R_2 = \{(a, a), (b, b), (c, c), (d, d), (e, e), (f, f), (g, g), (h, h), (i, i), (j, j), (k, k), (a, b), (a, c), (a, d), (b, c), (b, d), (c, d), (b, a), (c, a), (d, a), (c, b), (d, b), (d, c), (a, h), (b, h), (c, h), (h, a), (h, b), (h, c), (e, f), (e, g), (f, g), (f, e), (g, e), (g, f), (h, i), (i, h), (j, k), (k, j), (d, i), (i, d), (e, j), (e, k), (f, j), (f, k), (j, e), (k, e), (j, f), (k, f)\}$

但$R_1 \cup R_2$不是传递的，如：（c， h），（h，i）$\in R_1 \cup R_2$，而（c， i）$\notin R_1 \cup R_2$。　把$R_1 \cup R_2$扩张到传递性满足，就是（$R_1 \cup R_2$）*。

$R_1 \cap R_2 \leftrightarrow \pi_1 \cdot \pi_2$ □

$$= \{ \underset{A_1 \cap B_1}{\overline{abc}}, \underset{A_1 \cap B_2}{\overline{d}}, \underset{A_2 \cap B_2}{\overline{ef}}, \underset{A_4 \cap B_4}{\overline{g}}, \underset{A_3 \cap B_1}{\overline{h}}, \underset{A_3 \cap B_2}{\overline{i}}, \underset{A_4 \cap B_3}{\overline{jk}} \}$$

□凡是在π_1，π_2中有一个要求把它们拆开的就拆开。□

$$(R_1 \cup R_2)^* \leftrightarrow \{ \underset{A_1 \cup A_3 = B_1 \cup B_2}{\overline{abcdih}}, \underset{A_2 \cup A_4 = B_3 \cup B_4}{\overline{efgjk}} \}$$

例7：

$A=\{52$张扑克牌$\}$

$R_1=\{(a, b) \mid a$与b同花，a, b是扑克$\}$

$\leftrightarrow \pi_1$

$R_2=\{(a, b) \mid a$与b同点，a, b是扑克$\}$

$\leftrightarrow \pi_2$

$R_1 \cap R_2 \leftrightarrow \pi_1 \cdot \pi_2$把扑克分成52张单张

$(R_1 \cup R_2)^* \leftrightarrow \pi_1 + \pi_2$把52张扑克归为同一类

[定义]**平凡分划**：

集合A的最粗或最细的分划，称为**平凡分划**。

注：

（1）等价关系，用穷举法很难判别

（2）用关系图比较直观，明确

（3）用集合的分划来研究等价关系

作业

- § 9.4 – 20, 22, 28
- § 9.5 – 16, 56, 60, 64

§ 9.6: Partial Orderings

- A relation R on A is called a *partial ordering* or *partial order* iff it is reflexive, antisymmetric, and transitive.
  - We often use a symbol looking something like $\preccurlyeq$ (or analogous shapes) for such relations.
  - Examples: $\leq$, $\geq$ on real numbers, $\subseteq$, $\supseteq$ on sets.
  - Another example: the divides relation | on $\mathbf{Z}^+$.
    - Note it is not necessarily the case that either $a \preccurlyeq b$ or $b \preccurlyeq a$.
- A set A together with a partial order $\preccurlyeq$ on A is called a *partially ordered set* or *poset* and is denoted $(A, \preccurlyeq)$.

PARTIALLY ORDERED SETS (偏序集合）

- **Partial order**（偏序关系）:A relation *R* on a set *A* is called a *partial order* if *R* is reflexive（自反）, antisymmetric（反对称）, and transitive（传递）.
- **Partially ordered set**:The set *A* together with the partial order *R* is called a *partially ordered set*, or simply a *poset*, and we will denote this poset by (*A*, *R*).

Examples

- $(Z+, \leqslant)$
- $(Z+, \geqslant)$
- $(Z+, |)$
- $(Z+, <)$

linear order（线序关系）

- If (A, ≤) is a poset, the elements a and b are said to be
  - *Comparable（可比的）* if $a \le b$ or $b \le a$.
  - *incomparable (不可比的）* if neither $a \le b$ nor $b \le a$.
- If every pair of elements in a poset A is comparable, we say that A is a *linearly ordered set（全序集合）*, and the partial order is called a *linear order（全序关系或线序关系）*. We also say that A is a *chain（链）*.
  - $(Z^+, \leqslant)$
  - $(Z^+, |)$ is not a chain.

Well- ordered（良序关系）

- (S,≤) is a well-ordered set if it is a poset such that ≤ is a total ordering and every nonempty subset of S has a least element.

Product partial order
（乘积偏序）

If $(A, \leq)$ and $(B, \leq)$ are posets, then $(A \times B, \leq)$ is a poset, with partial order $\leq$ defined by

$$(a, b) \leq (a', b') \text{ if } a \leq a' \text{ in } A \text{ and } b \leq b' \text{ in } B.$$

- This ordering is called *product partial order*.

Lexicographic order
（词典顺序）

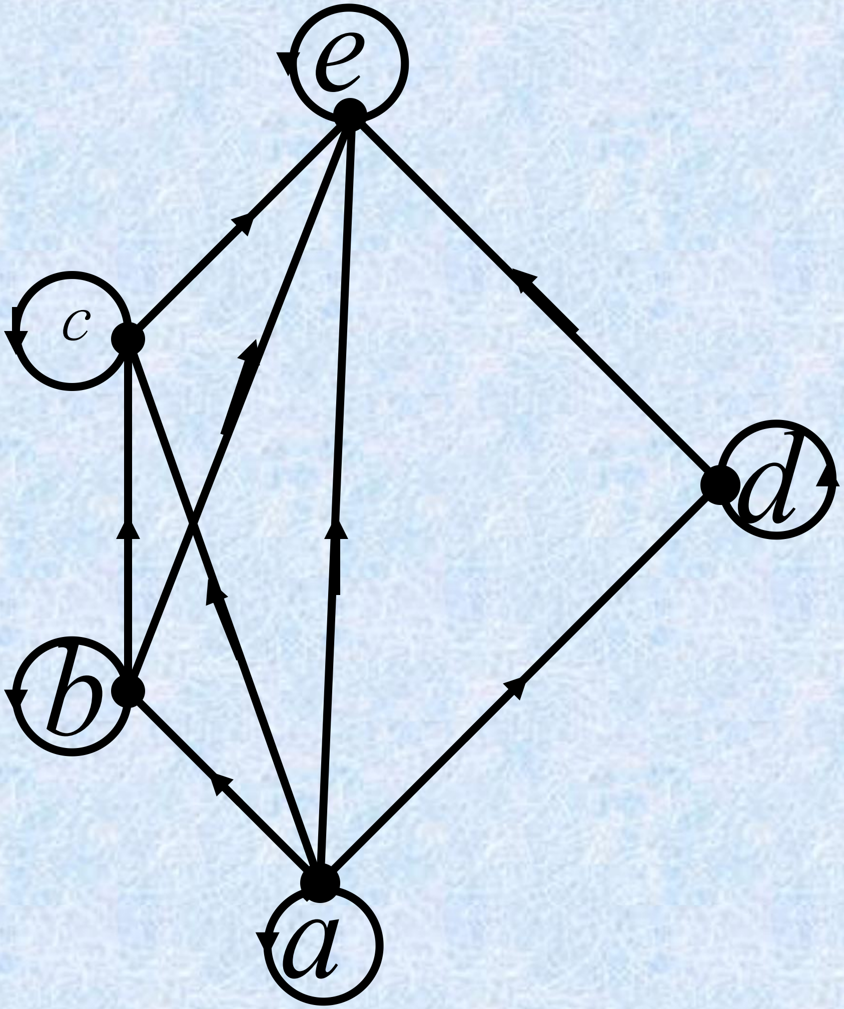
If $(A,\leq)$ and $(B,\leq)$ are poset, then $(A\times B,\prec)$ is a poset, with partial order $\prec$ defined by

$$(a,b)\prec(a',b') \text{ if } a<a' \text{ or if } a=a' \text{ and } b\leq b'.$$

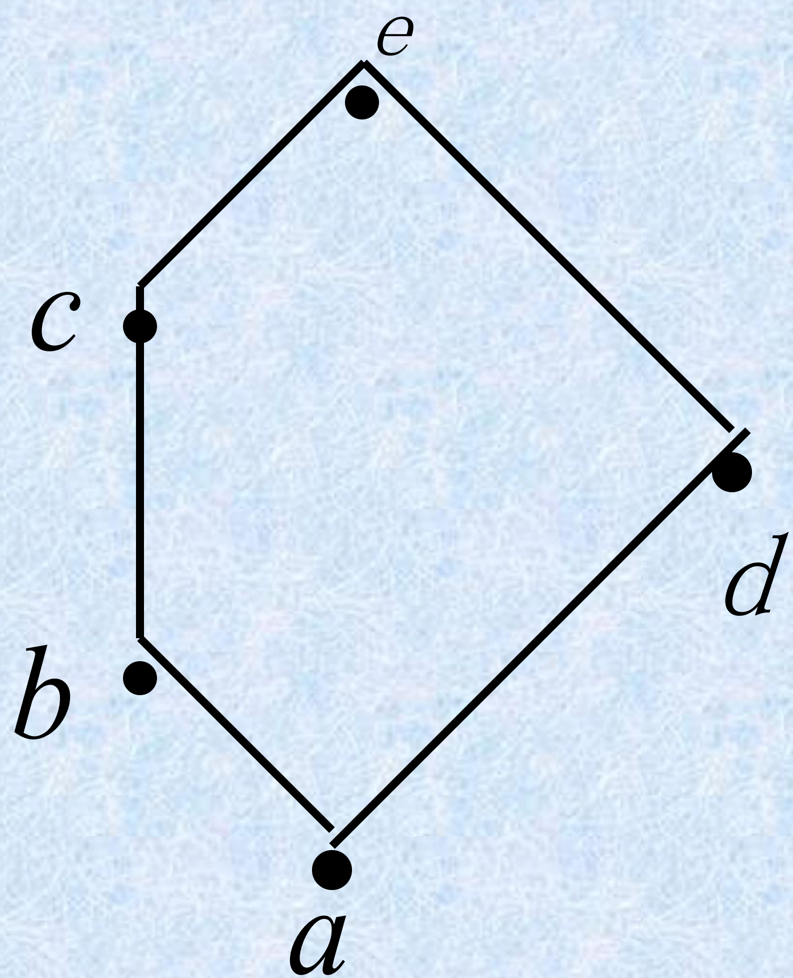
- This ordering is called *lexicographic*, or *"dictionary" order*.
- Extended to Cartesian products $A_1XA_2\ldots XA_n$
- Ordinary alphabet help $\prec$ helping helper $\nprec$ helping

Hasse Diagrams（哈斯图）

- Hasse Diagram is the digraph of a partial order on a finite set *A* that
  - deletes all self-cycles
  - eliminates all edges that are implied by the transitive property
  - draws the digraph with all edges pointing upward

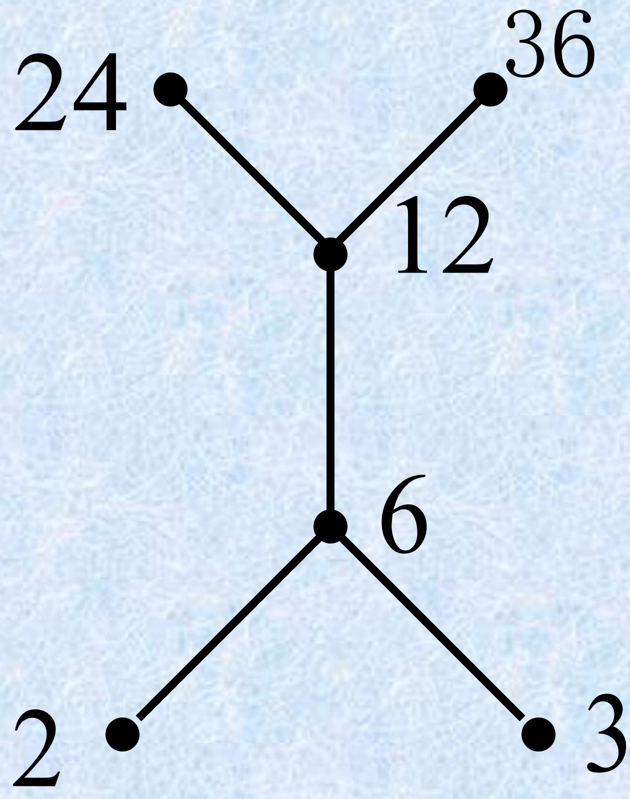


关系图



哈斯图

$A_3 = \{2,3,6,12,24,36\}$中，$\leq$ 为 A_3中的整除关系，$< A_3, \leq >$ 的哈斯图：



Extremal elements
（极值元素）

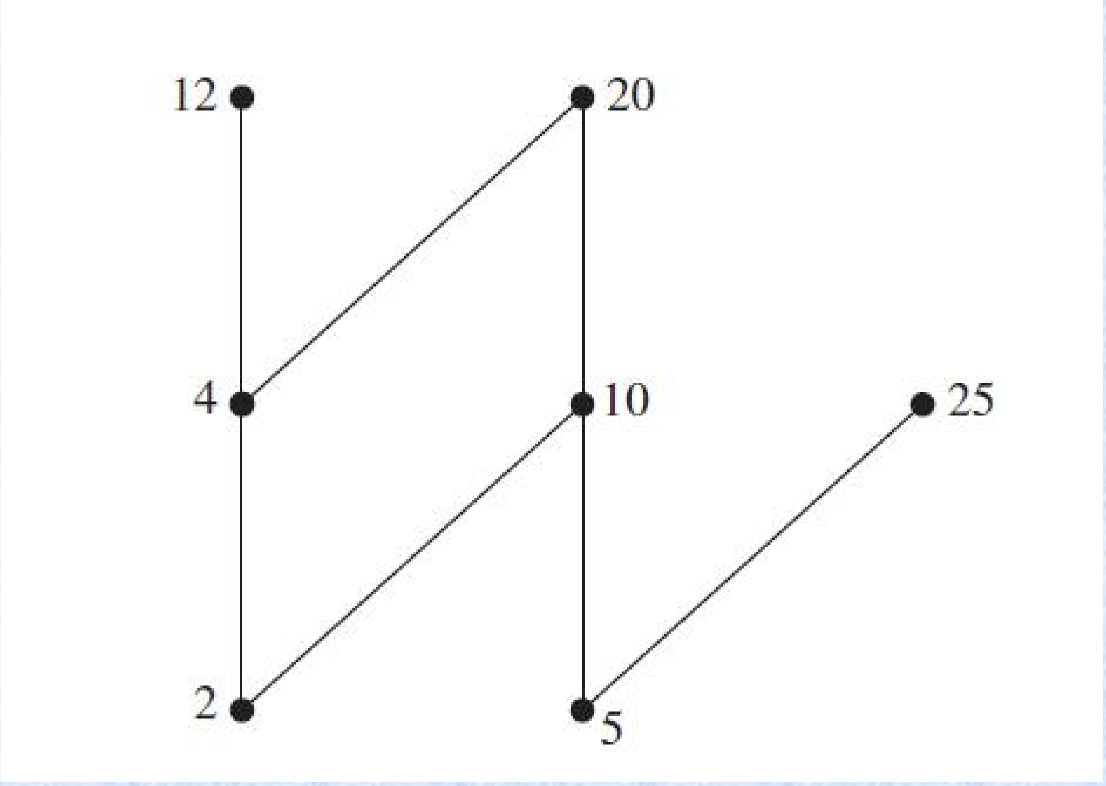
Extremal elements

- *maximal element*（极大元）
- *minimal element*（极小元）
- *greatest element*（最大元）
  - denoted by *I* and often called the *unit element*（单位元）
- *least element*（最小元）
  - denoted by *0* and often called the *zero element*（零元）

Maximal(minimal) element

- An element $a \in A$ is called a maximal element of A if there is no element c in A such that $a<c$.
- An element $b \in A$ is called a minimal element of A if there is no element c in A such that $c<b$.

Example 14 Which elements of the poset (2,4,5,10,12,20,25},|) are maximal ,and which are minimal?

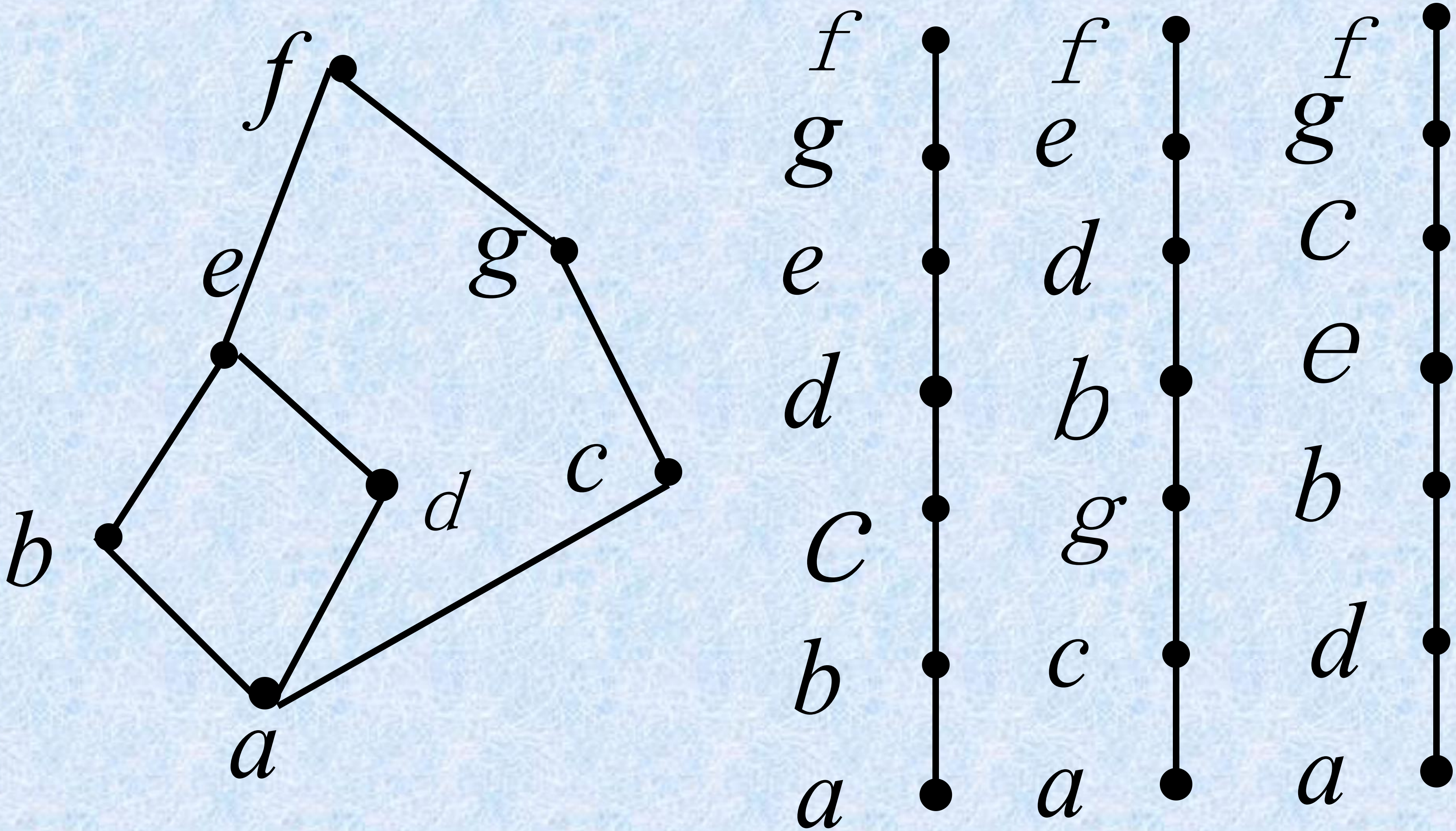


Theorem 1

- Let A be a finite nonempty poset with partial order ≤, then A has at least one maximal element and at least one minimal element.

Topological Sorting
（拓扑排序）

- If *A* is a poset with partial order ≤, we sometimes need to find *a linear order* $\prec$ for the set *A* that will merely be an extension of the given partial order in the sense that if $a \leq b$, then $a \prec b$.
- The process of constructing a linear order such as $\prec$ is called *topological sorting*


f
e
g
b
d
c
a
f g e d c b a
f e d b g c a
f g c e b d a

- Find a compatible total ordering for the poset
- $(\{1,2,4,5,12,20\},|)$

- Find an ordering of the tasks of a software project if the Hasse diagram for the tasks of the project is as shown. P581 64

SORT Algorithm

- For finding a topological sorting of a finite poset (A, ≤).
- Step1 Choose a minimal element a of A.
- Step2 Make a the next entry of SORT and replace A with A-{a}.
- Step3 Repeat steps 1 and 2 until A={}.
- End of Algorithm

Greatest (least) element

- An element a A is called a **greatest element** of A if $x \leq a$ for all $x \in A$.
- An element a A is called a **least element** of A if $a \leq x$ for all $x \in A$.
- The greatest element of a poset, if it exists, is denoted by *I*, and is often called the **unit** element. Similarly, the least element is called **zero** element.

定义 偏序集 $<A,\leq>$,且 B 是 A 的子集,若有 $b \in B$, 对所有 $x \in B$ 有 $x \leq b$, $(b \leq x)$. 则称 b 为 $<B,\leq>$ 的最大元.

(最小元)

y 为 B的最小元,则它“ 小于”

y 为 B的最大元,则它“ 大于”

B 中所有其他元素

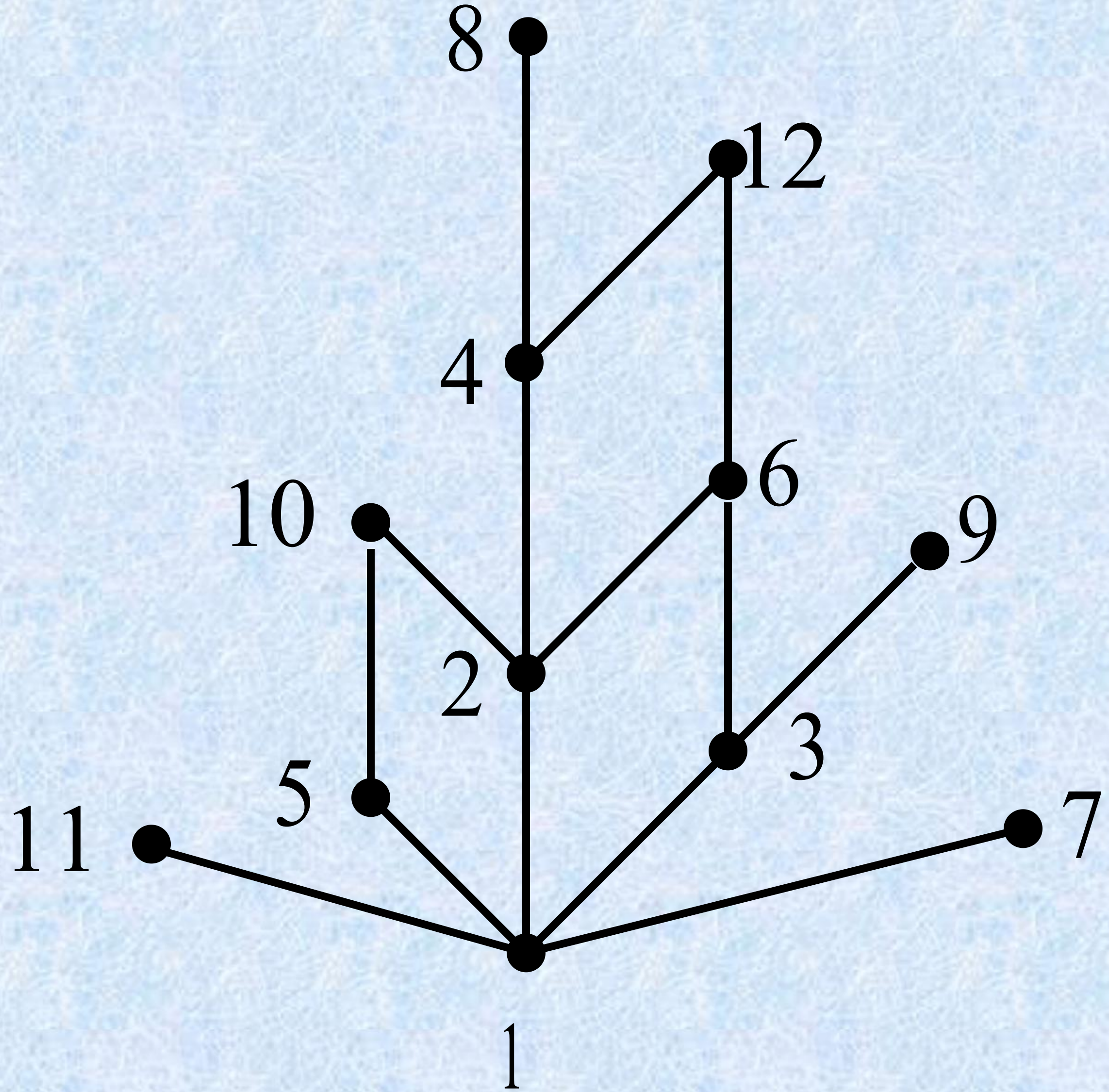
y 为 B的极小元, 若在B 中没

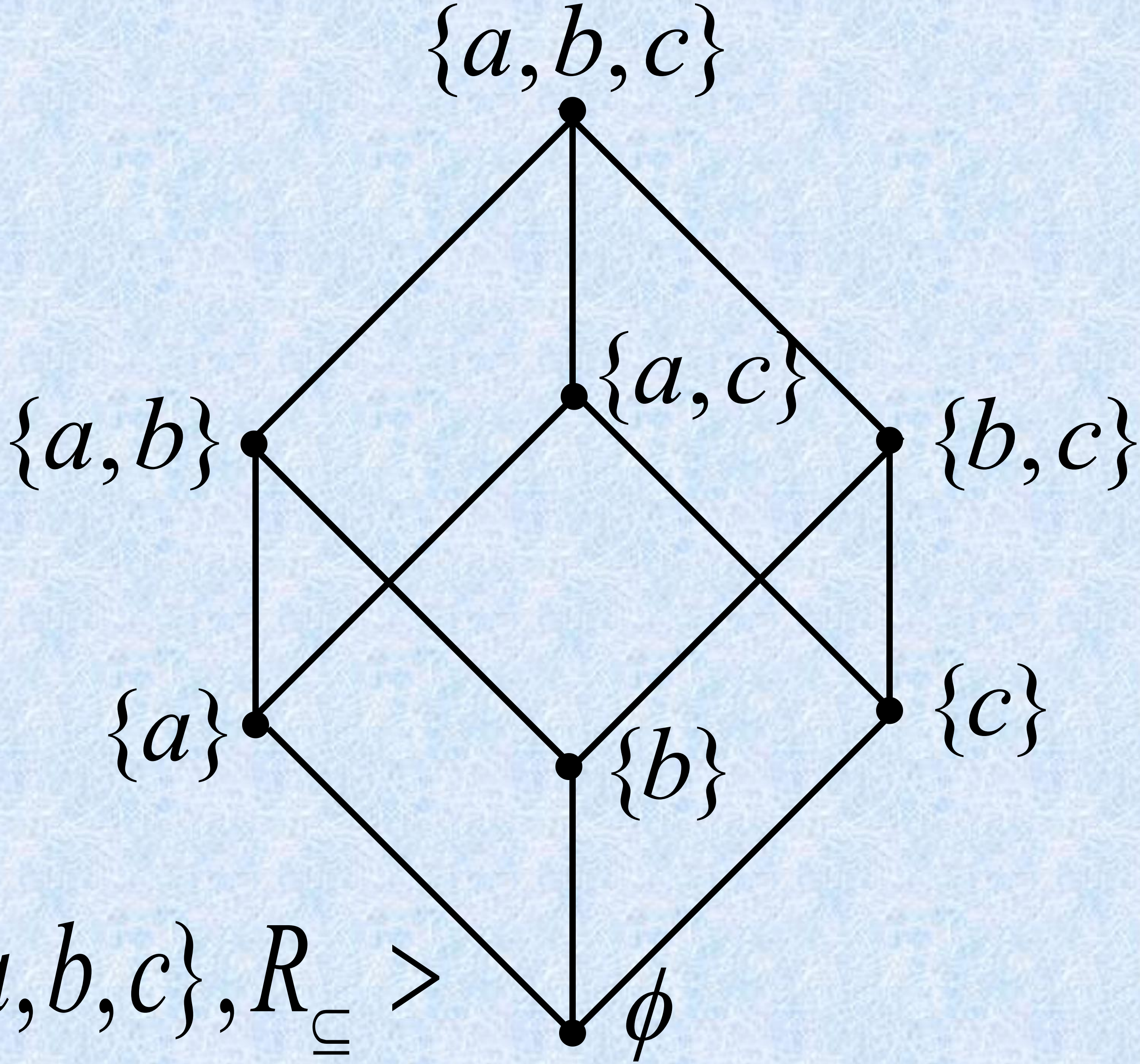
y 为 B的极大元.

有比 y 小的其他元存在.

大

$$<\{1,2,\cdots,12\},R_{\text{整除}}>$$

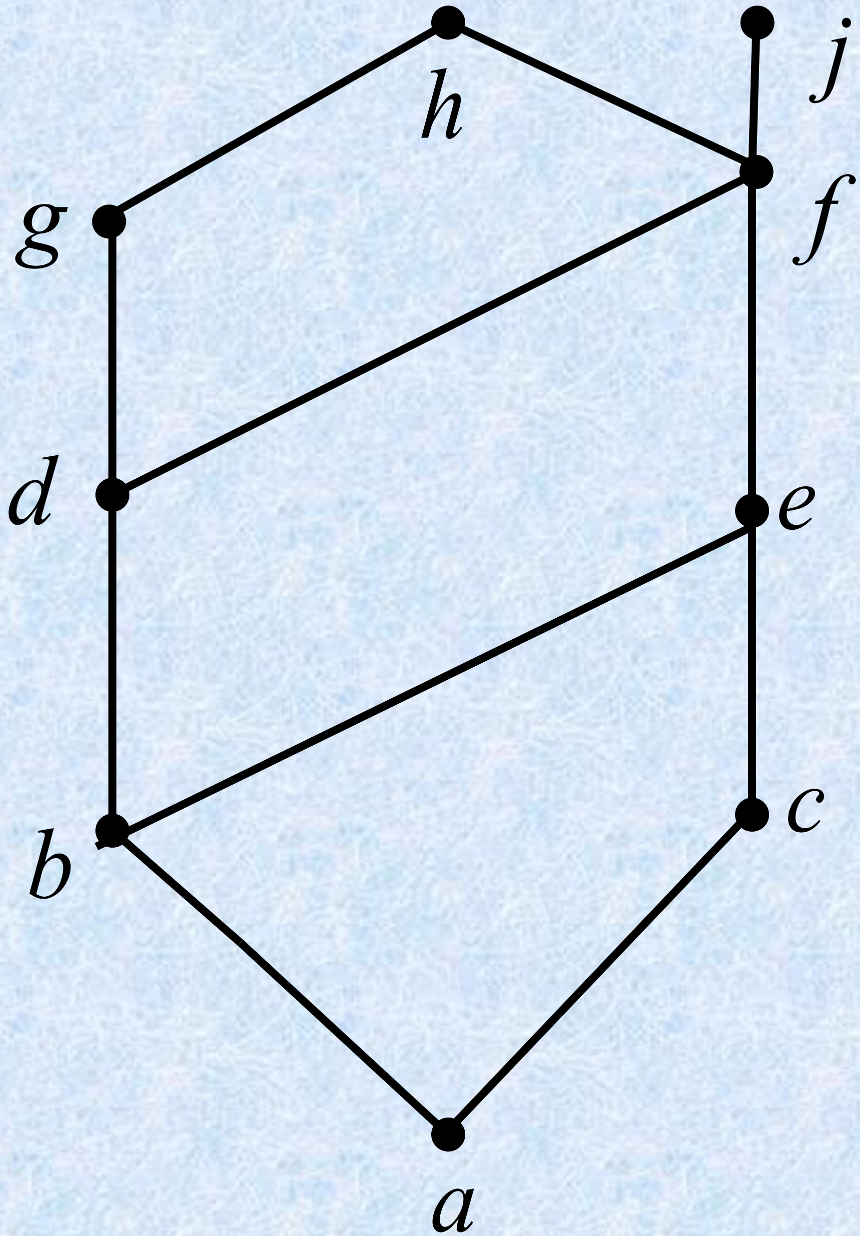




$<P\{a,b,c\},R_{\subseteq}>$

Lub（最小上界）Glb（最大下界）

- Consider a poset and a subset B of A.
  - An element $a \in A$ is called a *upper bound*（上界） of B if $b \leq a$ for all $b \in B$.
  - An element $a \in A$ is called a *lower bound*（下界）of B if $a \leq b$ for all $b \in B$.
  - An element $a \in A$ is called a *least upper bound* (LUB) of B if a is an upper bound of B and $a \leq a'$, whenever a' is an upper bound of B.
  - An element $a \in A$ is called a *greatest lower bound* (GLB) of B if a is an lower bound of B and $a' \leq a$, whenever a' is an lower bound of B.

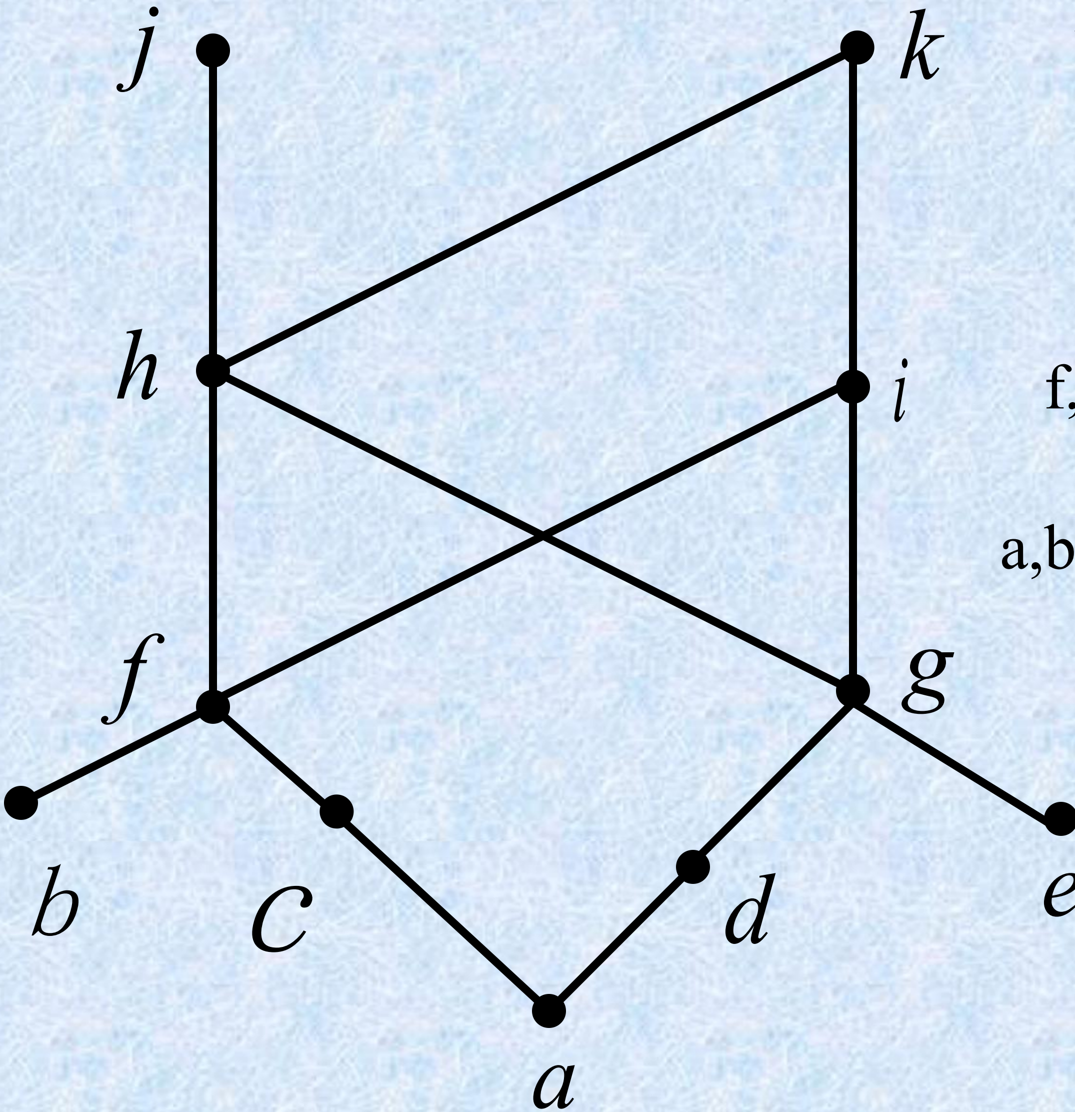


Find the lower and upper bounds of the subsets {a,b,c}, {j,h}, and {a,c,d,f}

Find the greatest lower bound and least upper bound of the subsets {b,d,g},if they exist.

Example

Find the greatest lower bound and the least upper bound of the sets {3,9,12} and {1,2,4,5,10},if they exist, in the poset{Z^+,|}.



$$B = \{a, b, c, d, e, f, g\}$$

h,i 为B的上界

f,g 为B’={j,h,k,i}的下界

a,b,c,d,e 也为B’={j,h,k,i}的下界

a是{f,g,h,i,j,k}的下界

b,c,d,e不是{f,g,h,i,j,k}的下界

Theorem 1,2,3

- Let (A, $\leqslant$) be a poset
  - If A is a finite nonempty, then A has at least *one* maximal element and at least *one* minimal element.
  - A has at most *one* greatest element and at most *one* least element.
  - A subset B of A has at most *one* LUB and at most *one* GLB.

Theorem 4

- Suppose that (A, ≤) and (A', ≤') are isomorphic posets under the isomorphism f: $A \rightarrow A'$.
  - If a is a maximal (minimal) element of (A, ≤), then $f(a)$ is a maximal (minimal) element of (A', ≤').
  - If a is a greatest (least) element of (A, ≤), then $f(a)$ is a greatest (least) element of (A', ≤').
  - If a is an upper bound (lower bound, least upper bound, greatest lower bound) of (A, ≤), then $f(a)$ is an upper bound (lower bound, least upper bound, greatest lower bound) of (A', ≤').
  - If every subset of (A, ⩽) has a LUB (GLB), then every subset of (A', ⩽') has a LUB (GLB).

Lattices（格）

lattices

- Definition lattices sublattice
- Isomorphic lattices
- Properties of lattices
- Special types of lattices
- bounded distributive complemented
- modular

Lattice

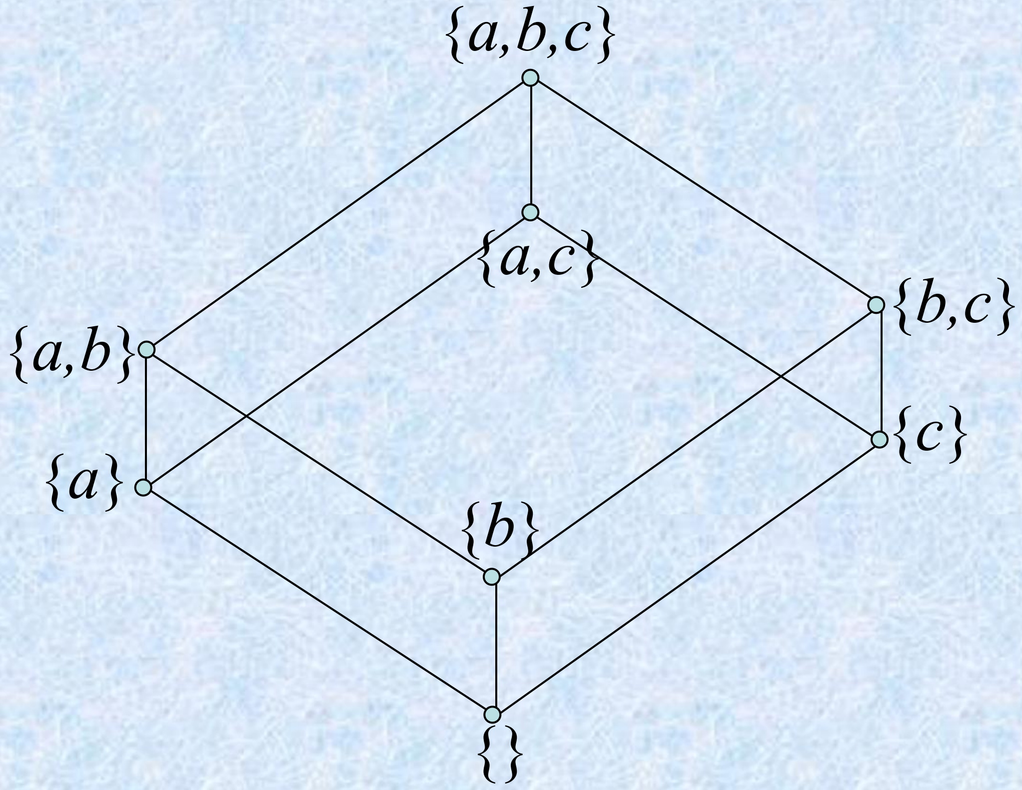
- A *lattice* is a poset in which every subset {a,b} consisting of two elements has a least upper bound and a greatest lower bound.
  - denote LUB({a,b}) by $a \vee b$ and call it the *join*（*并*） of a and b.
  - denote GLB({a,b}) by $a \wedge b$ and call it the *meet*（*交*） of a and b.

Example

- Let S be a set and let $L = P(S)$.
  - As we have seen, $\subseteq$, containment, is a partial order on L. Let A and B belong to the poset $(L, \subseteq)$.
- $A \vee B$ is the set $A \cup B$.
  - To see that, note that $A \subseteq A \cup B$, $B \subseteq A \cup B$, and if $A \subseteq C$ and $B \subseteq C$, then it follows that $A \cup B \subseteq C$.
- Similarly, $A \wedge B$ is the set $A \cap B$.
- Thus, L is a lattice.

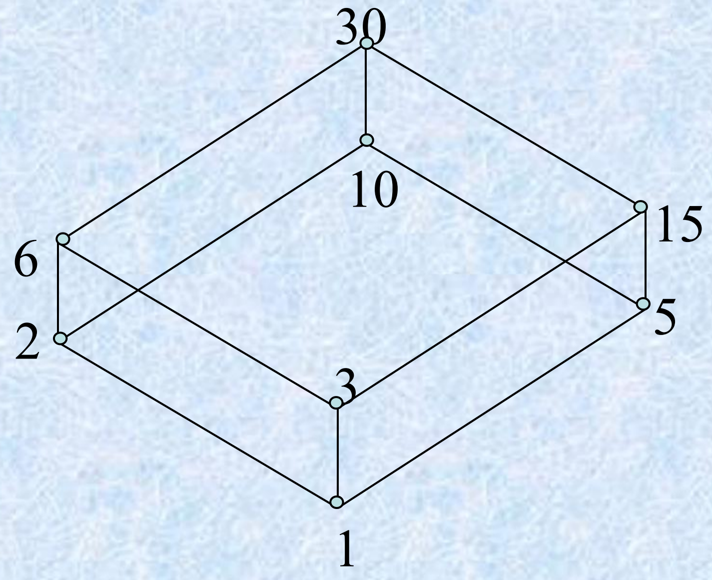
Example

- $S=\{a,b,c\}$

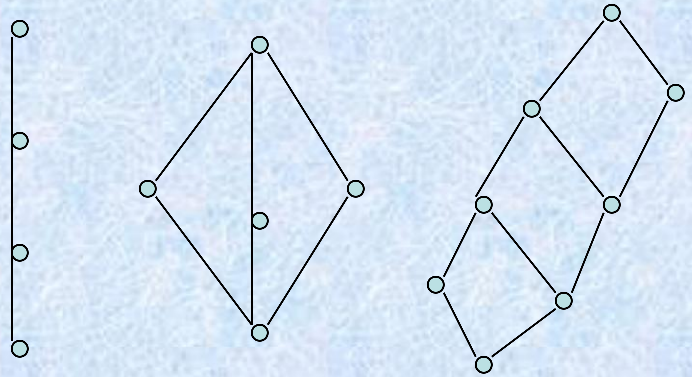


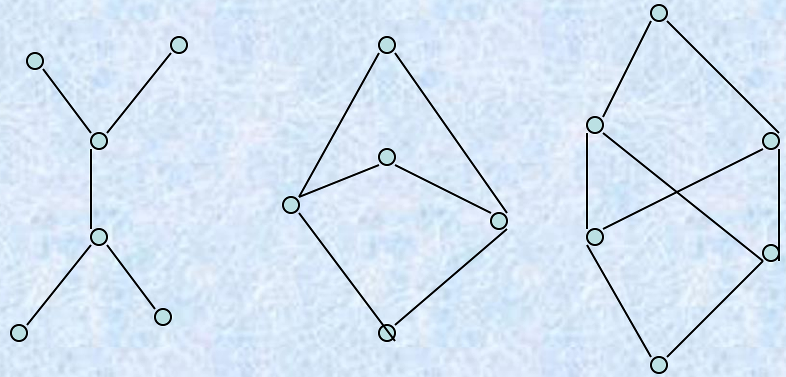
Example

- Let n be a positive integer and D_n be the set of all positive divisors of n.
- D_{30}={1,2,3,5,10,15,30}
  - $a \vee b$=LCM(a,b)
  - $a \wedge b$=GCD(a,b)



例1:判断是否为格



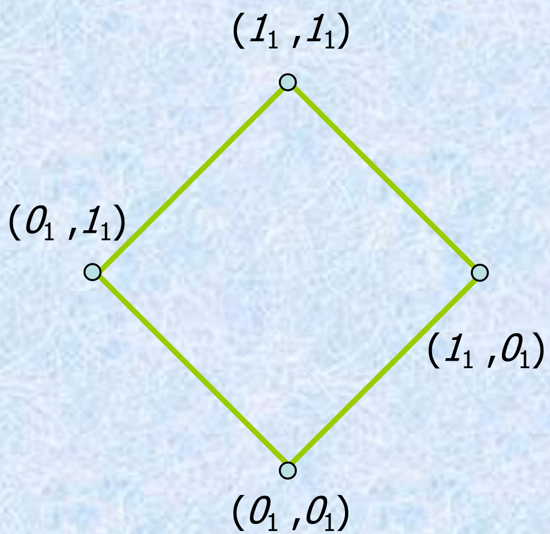


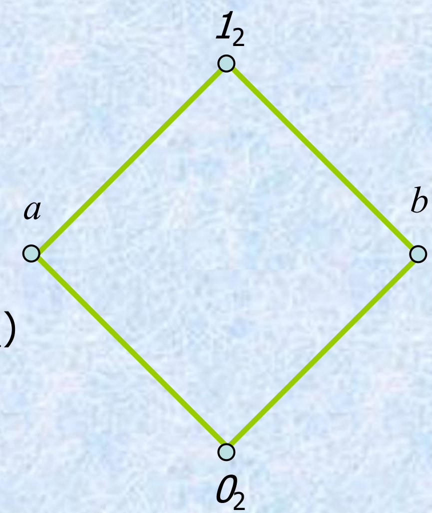
格

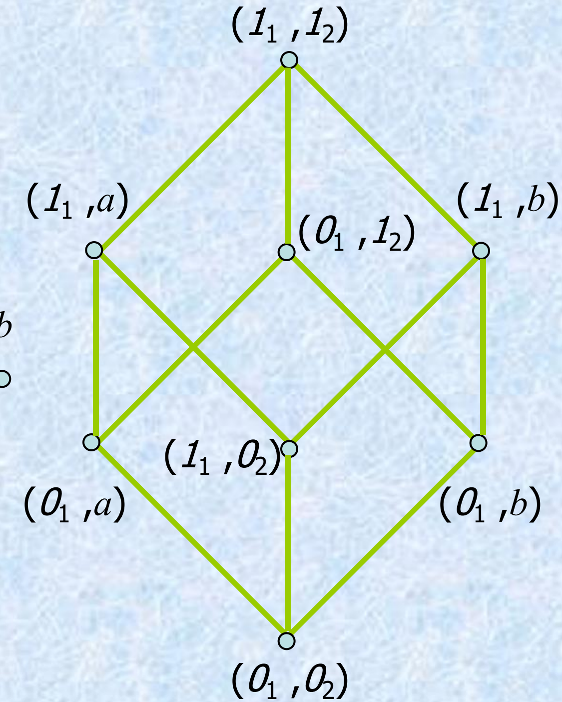
不是格的偏序集合

Example

L_1 $L_{2'} = L_1 \times L_1$ L_2 $L = L_1 \times L_2$





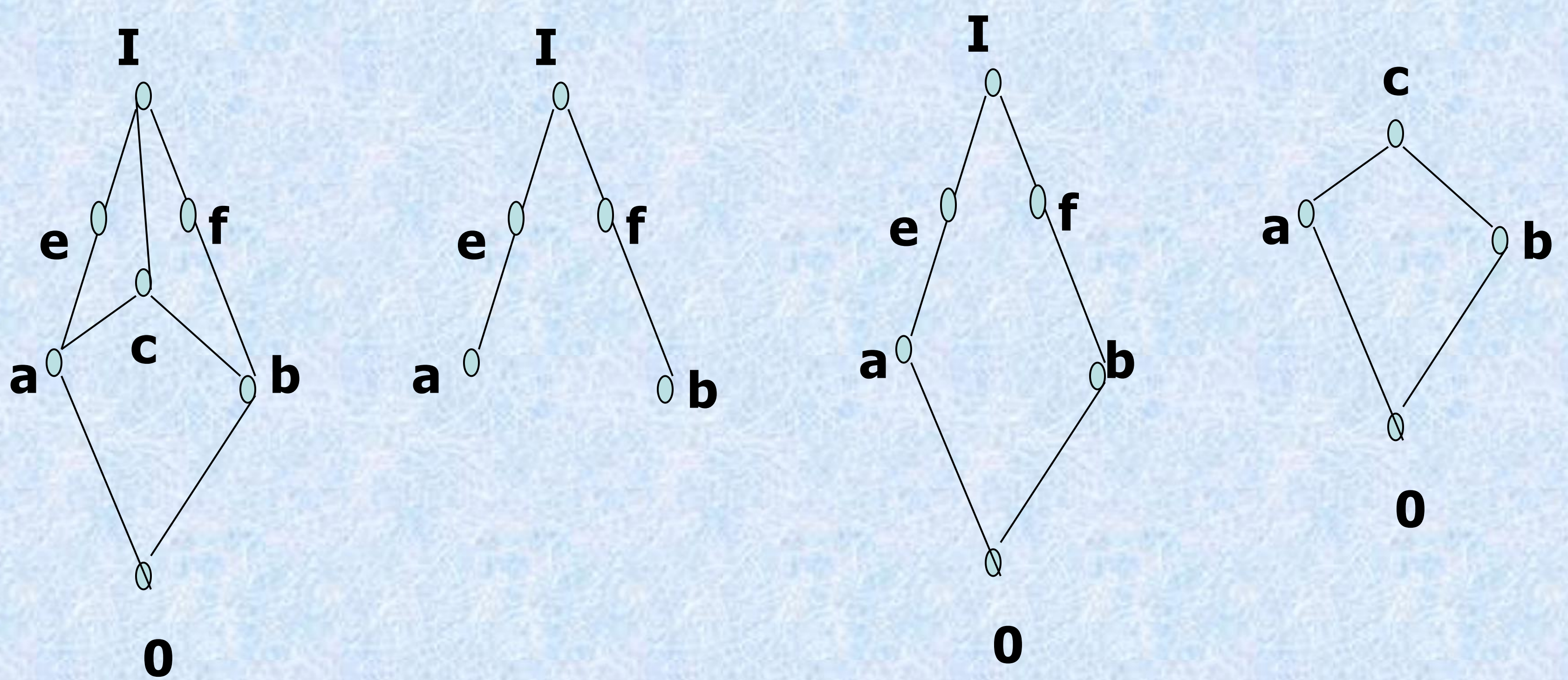


Sublattice（子格）

- Let $(L, \leq)$ be a lattice. A nonempty subset S of L is called a *sublattice* of L
  - if $a \vee b \in S$ and $a \wedge b \in S$ whenever $a \in S$ and $b \in S$.

Sublattice

- $(D_n, |)$ is a sublattice of $(Z^+, |)$



Isomorphic Lattices（同构格）

- If f: $L_1 \rightarrow L_2$ is an isomorphism from the poset $(L_1, \leq_1)$ to the poset $(L_2, \leq_2)$
  - L_1 is a lattice if and only if L_2 is a lattice.
  - If a and b are elements of L_1, then
    - $f(a \wedge b) = f(a) \wedge f(b)$
    - $f(a \vee b) = f(a) \vee f(b)$.
- If two lattices are isomorphic, as posets, we say they are *Isomorphic Lattices*.

Properties of Lattices

- Recall the meaning of $a \vee b$ and $a \wedge b$:
  - $a \le a \vee b$ and $b \le a \vee b$; $a \vee b$ is an *upper bound* of a and b.
  - If $a \le c$ and $b \le c$, then $a \vee b \le c$; $a \vee b$ is the *least upper bound* of a and b.
  - $a \wedge b \le a$ and $a \wedge b \le b$; $a \wedge b$ is an *lower bound* of a and b.
  - If $c \le a$ and $c \le b$, then $c \le a \wedge b$; $a \wedge b$ is the *greatest lower bound* of a and b.

Theorem 2

- Let L be a lattice. Then for every a and b in L,
  - $a \vee b = b$ if and only if $a \leq b$.
  - $a \wedge b = a$ if and only if $a \leq b$.
  - $a \wedge b = a$ if and only if $a \vee b = b$

Proof of $a \vee b = b$ if and only if $a \leq b$.

- $\Rightarrow$ only if
  - Suppose $a \vee b = b$, so $a \vee b \leq b$ (reflexive)
  - $a \vee b$ is an upper bound of a and b, so $a \leq a \vee b$
  - So, $a \leq b$ (transitive)
- $\Leftarrow$ if
  - Suppose $a \leq b$
  - $b \leq b$ (reflexive), b is an upper bound of a and b
  - So $a \vee b \leqslant b$ (by definition of lub)
  - But, $b \leqslant a \vee b$ (by definition of lub)
  - Therefore, $a \vee b = b$ (antisymmetric)

Theorem 3

- Let L be a lattice. Then
  - Idempotent Properties（等幂律）
    - (a) $a \vee a = a$
    - (b) $a \wedge a = a$
  - Commutative Properties（交换律）
    - (a) $a \vee b = b \vee a$
    - (b) $a \wedge b = b \wedge a$
  - Associative Properties（结合律）
    - (a) $a \vee (b \vee c) = (a \vee b) \vee c$
    - (b) $a \wedge (b \wedge c) = (a \wedge b) \wedge c$
  - Absorption Properties（吸收律）
    - (a) $a \vee (a \wedge b) = a$
    - (b) $a \wedge (a \vee b) = a$

Proof of 4. (a)

- Since $a \wedge b \leqslant a$ and $a \leqslant a$, we see that a is an upper bound of $a \wedge b$ and a; so $a \vee (a \wedge b) \leqslant a$. On the other hand, by the definition of LUB, we have $a \leqslant a \vee (a \wedge b)$, so $a \vee (a \wedge b) = a$.
  - Q.E.D
- It follows from property 3 that we can write $a \vee (b \vee c)$ and $(a \vee b) \vee c$ merely as $a \vee b \vee c$, and similarly for $a \wedge b \wedge c$.

Theorem 4

- Let L be a lattice. Then for every a, b, and c in L,
- If $a \leq b$, then
  - $a \vee c \leq b \vee c$.
  - $a \wedge c \leq b \wedge c$.
- $a \leq c$ and $b \leq c$ if and only if $a \vee b \leq c$.
- $c \leq a$ and $c \leq b$ if and only if $c \leq a \wedge b$.
- If $a \leq b$ and $c \leq d$, then
  - $a \vee c \leq b \vee d$.
  - $a \wedge c \leqslant b \wedge d$.

Special Types of Lattices

- A lattice *L* is said to be *bounded*（*有界的*）if it has a greatest element *I* and a least element *0*

Example 15

- The lattice P(S) of all subsets of a set S, as defined in Example 1, is bounded. Its greatest element is S and its least element is $\varnothing$
- If L is a bounded lattice, then for all $a \in A$,
  - $0 \leqslant a \leqslant I$
  - $a \vee 0 = a$
  - $a \wedge 0 = 0$
  - $a \vee I = I$
  - $a \wedge I = a$

Theorem 5

- Let $L = \{a_1, a_2, \ldots, a_n\}$ be a finite lattice. Then L is bounded.
- **Proof**
  - The greatest element of L is $a_1 \vee a_2 \vee \ldots \vee a_n$,
  - and its least element is $a_1 \wedge a_2 \wedge \ldots \wedge a_n$

Distributive lattice（分配格）

- A lattice is called *distributive* if for any elements *a*, *b*, and *c* in *L* we have the following distributive properties.
  - $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$
  - $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
- If L is not distributive, we say that *L* is *nondistributive*

nondistributive lattices

- The lattice shown in Figure 6.44 are nondistributive.

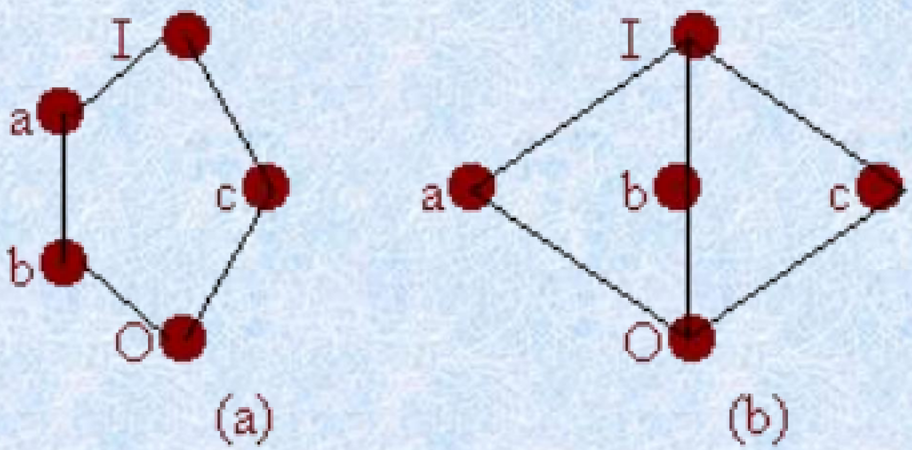


Figure 6.44

- $a \wedge (b \vee c)$ =a
  $(a \wedge b) \vee (a \wedge c)$=b

- $a \wedge (b \vee c)$ =a
  $(a \wedge b) \vee (a \wedge c)$=0

Theorem 6

- A lattic *L* is nondistributive if and only if it contains a sublattice that is isomorphic to one of the above two lattices.
- Proof:
  - omitted

Complement（补元）

- Let L be a bounded lattice with greatest element I and least element 0, and let $a \in L$. An element a' L is called a *complement of a* if
  - $a \vee a' = I$ and $a \wedge a' = 0$
- Note that
  - $0' = I$ *and* $I' = 0$.
- An element a in a lattice need not have a complement, and it may have more than one complement.

Example 20

- The lattices in Figure 6.44 each have the property that every element has a complement. The element *c* in both cases has two complements, *a* and *b*.

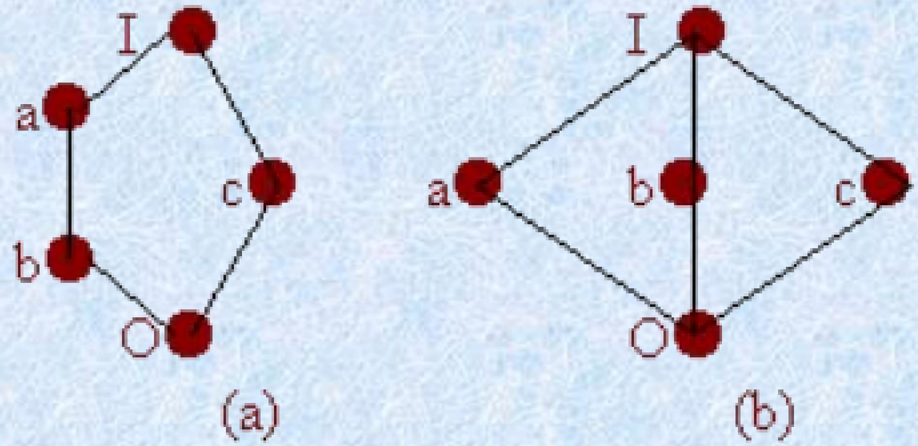
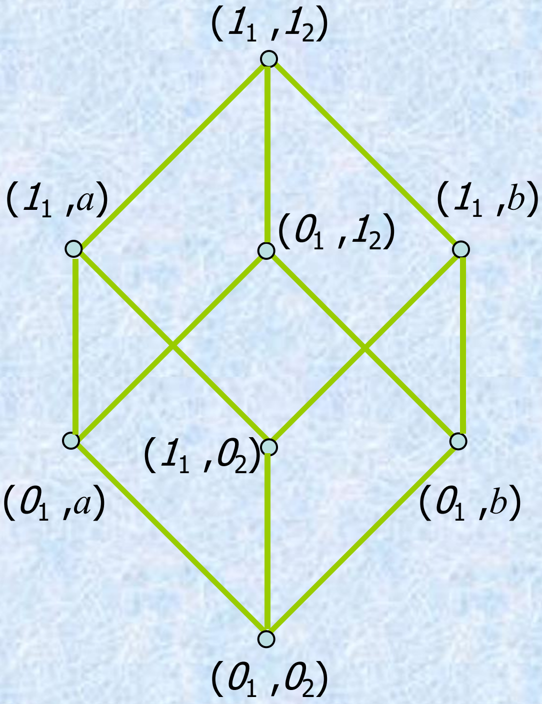


Figure 6.44

Theorem 7

- Let *L* be a bounded distributive lattice. If a complement exists, it is unique.



Proof

- Let a' and a'' be complements of the element $a \in L$. Then
  - $a \vee a' = I$, $a \vee a'' = I$, $a \wedge a' = 0$, $a \wedge a'' = 0$.
- Using the distributive laws, we obtain
  - $a' = a' \vee 0 = a' \vee (a \wedge a'') = (a' \vee a) \wedge (a' \vee a'') = I \wedge (a' \vee a'') = a' \vee a''$
- Also
  - $a'' = a'' \vee 0 = a'' \vee (a \wedge a') = (a'' \vee a) \wedge (a'' \vee a') = I \wedge (a'' \vee a') = a' \vee a''$
- Hence $a' = a''$.

Complemented（有补格）

- A lattice is called *complemented* if it is bounded and if every element in *L* has a complement.
- Example 23 In this case, the complements are not unique.

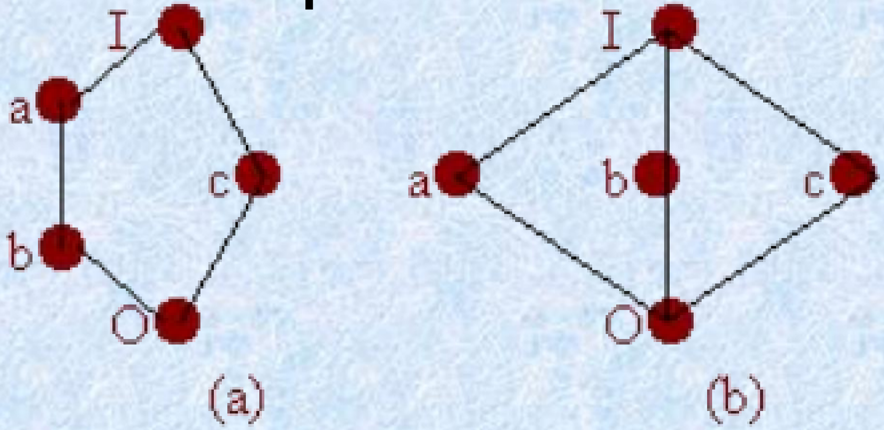


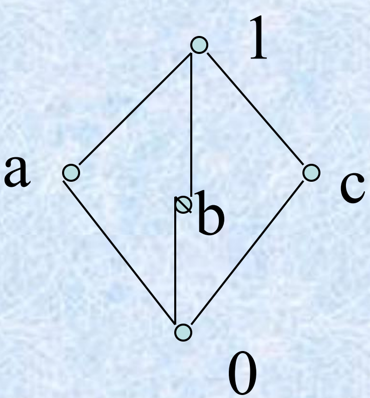
Figure 6.44

Modular 模格

1.定义： 若$\forall a,b,c \in \langle L,\leqslant\rangle$，

$$a \leqslant c \Rightarrow a \vee (b \wedge c) = (a \vee b) \wedge c$$

称$\langle L,\leqslant\rangle$是模格。



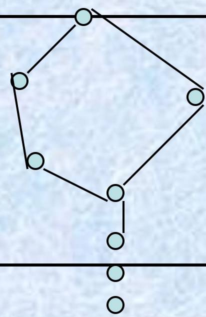
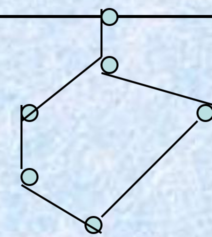
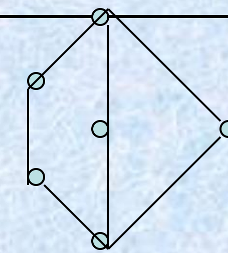
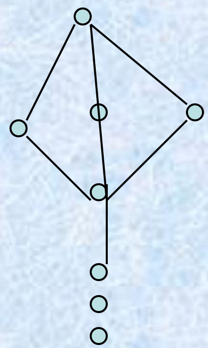
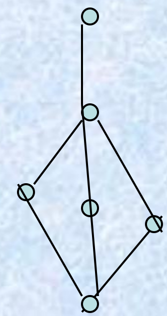
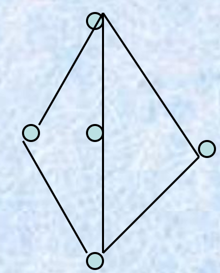
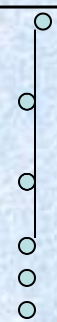
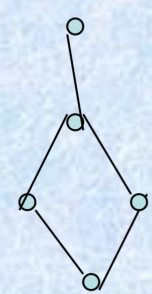
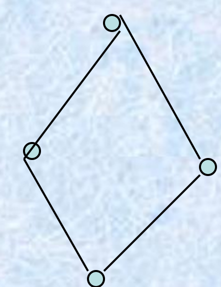
例1： 它是模格，但不是分配格

$a \vee (b \wedge c) = a \vee 0 = a$

$(a \vee b) \wedge (a \vee c) = 1 \wedge 1 = 1$

lattices

- Lattices
  - modular
    - distributive
    - nondistributive
  - nonmodular

<table>
<tr><td colspan="2" rowspan="2"></td><td rowspan="2">非有界格</td><td colspan="2">有界格</td></tr>
<tr><td>非有补格</td><td>有补格</td></tr>
<tr><td colspan="2">非模 格</td><td></td><td></td><td></td></tr>
<tr><td rowspan="2">模格</td><td>非分配格</td><td></td><td></td><td></td></tr>
<tr><td>分配格</td><td></td><td></td><td></td></tr>
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dual

- Let R be a partial order on a set A, and let R^{-1} be the inverse relation of R, the R^{-1} is also a partial order.
- The poset(A,R^{-1}) is called the dual of the poset(A,R), and the partial order R^{-1} is called the dual of the partial order R.

Quasiorder(拟序关系）

- *Quasiorder:* A relation *R* on a set *A* is called *quasiorder* if it is *transitive* and *irreflexive*.
  - Example:
    - (P(S),$\subset$)

作业

- § 9.1 – 42, 49, 50
- § 9.2 – 8, 20
- § 9.3 – 14, 32
- § 9.4 – 20, 22, 28
- § 9.5 – 16, 56, 60, 64
- § 9.6 – 12, 28, 36, 44